Exploring a Codebase

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1 Glossary - Summary

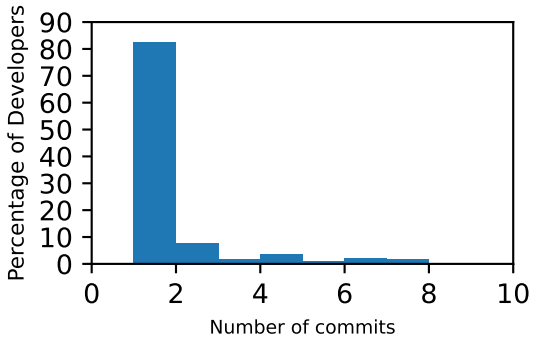
- A founder developer starts in the first six months of a project.
- A Later joiner developers start after six months.
- Transient developers have nine (9) or less commits.
- Moderate developers have more than nine (9) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

2 Repository Summary Table

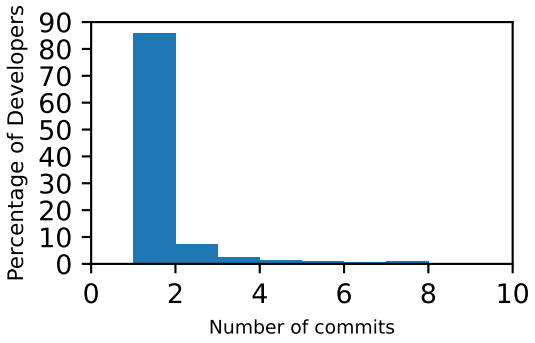
Table 1: Summary of thirty-one open-source repositories identified from GitHub Sustained later joiner developers who joined the project after six months and contributed at least 50 commits over a period of 250 days or more. Five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Later Joiner	Moderate Later Joiner	Sustained Later Joiner	Commit	Start	End
1	acs-aem-commons	3	0	0	189	0	0	1191	17-07-2013	Ongoing
3	activiti	6	7	6	210	22	5	9836	18-06-2010	Ongoing
5	airbyte-platform	16	2	1	454	35	6	7002	04-08-2020	27-09-2023
7	ambari	0	3	0	127	34	15	4357	22-09-2011	Ongoing
8	antlr4	3	0	0	293	0	0	3847	05-09-2011	Ongoing
10	automq	4	0	0	300	11	3	1889	10-08-2011	21-04-2017
11	avro	1	0	0	333	0	0	3339	15-04-2009	Ongoing
12	azure-sdk-for-java	8	0	0	11	0	0	1147	13-10-2011	08-01-2014
13	beam	57	0	0	38	0	0	6561	11-07-2014	20-05-2016
14	buck	17	6	6	524	92	21	15889	21-03-2013	05-05-2022
15	camel	1	4	1	1133	76	13	37463	19-03-2007	Ongoing
17	cas	2	0	0	305	0	0	46401	01-11-2010	Ongoing
18	checkstyle	0	0	1	359	45	9	17497	22-06-2001	Ongoing
19	commons-lang	4	0	0	202	0	0	3389	19-07-2002	Ongoing
20	cxf	6	6	3	201	17	3	8781	29-04-2008	Ongoing
21	intellij-community	0	3	7	334	103	84	104964	11-11-2004	18-04-2018
23	calcite	2	0	0	13	0	0	6211	20-04-2012	24-02-2014
24	dolphinscheduler	15	0	0	464	0	0	13029	29-03-2019	Ongoing
25	dubbo	4	0	0	176	0	0	4180	20-10-2011	07-04-2019
26	guava	0	1	0	400	10	3	9708	15-09-2009	Ongoing
28	jenkins	5	1	1	831	45	4	31546	05-11-2006	Ongoing
29	keycloak	3	0	0	1186	0	0	22954	02-07-2013	Ongoing
30	kubernetes-client	7	0	0	290	0	0	23224	09-07-2015	Ongoing
31	minecraftforge	2	0	0	493	0	0	20436	14-07-2011	Ongoing
32	openapi-generator	2	0	0	2646	0	0	30837	19-07-2011	Ongoing
33	openmrs-core	0	1	0	385	29	3	15769	03-05-2006	Ongoing
35	plugin-hub	18	0	0	347	0	0	8395	20-12-2019	Ongoing
36	presto	0	1	3	685	78	18	31858	09-08-2012	Ongoing
37	quarkus	2	3	5	820	29	8	82652	14-06-2018	Ongoing
38	runelite	0	1	0	673	0	0	59684	01-12-2014	Ongoing
39	selenium	5	3	0	709	20	8	23522	03-11-2004	Ongoing
	Total	193	42	34	15131	646	203	657558		

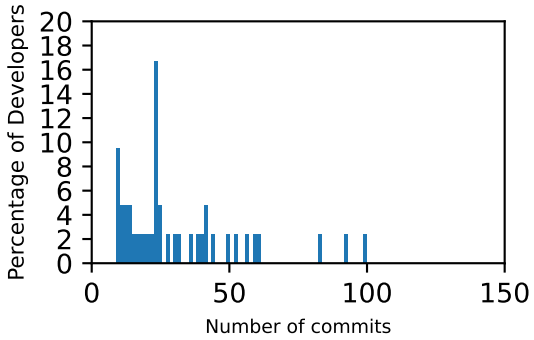
3 Total Commits By Developer - Developer Commits



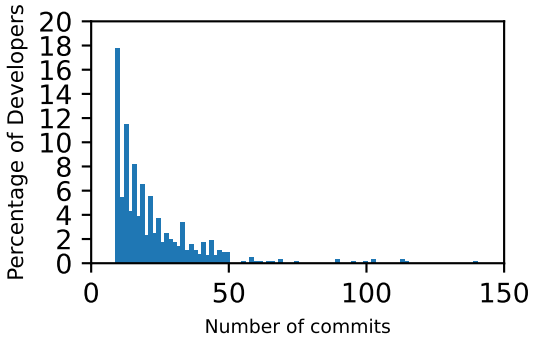
(a) Transient founder (n=193, y-axis max=82.4, x-axis max=8).



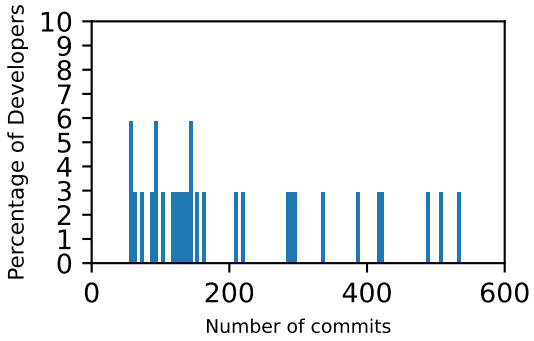
(b) Transient later joiner (n=15099, y-axis max=85.9, x-axis max=8).



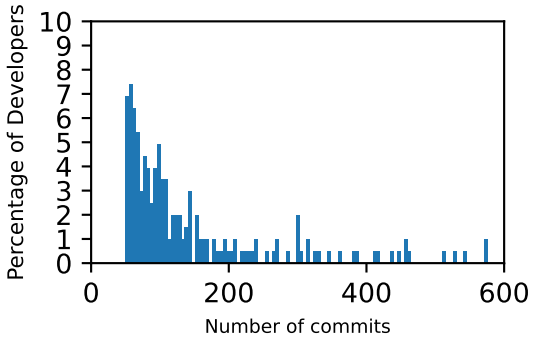
(c) Moderate founder (n=42, y-axis max=16.7, x-axis max=100).



(d) Moderate later joiner (n=646, y-axis max=17.8, x-axis max=304).



(e) Sustained founder (n=34, y-axis max=5.9, x-axis max=1255).

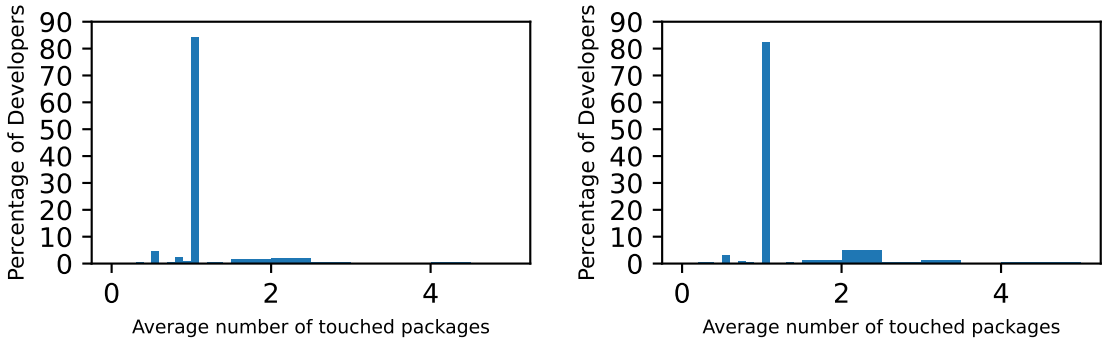


(f) Sustained later joiner (n=203, y-axis max=7.4, x-axis max=1062).

Figure 1: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 31 repositories sampled from GitHub.

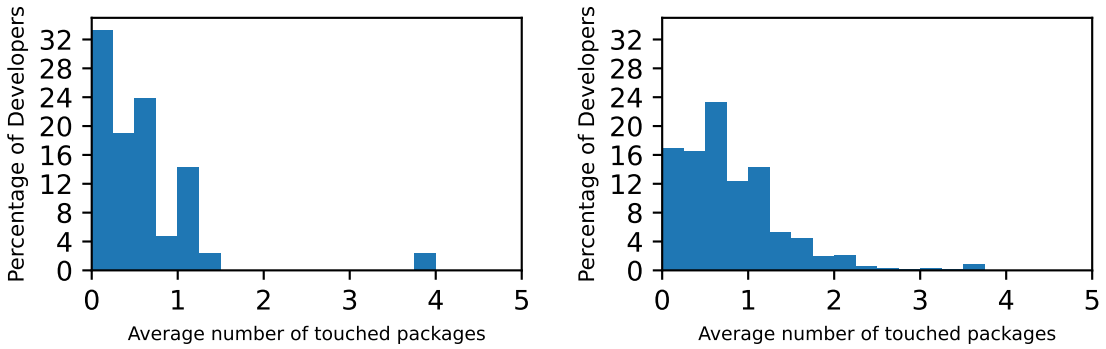
4 Histogram Components Touched By Developer - For components with total

4.1 Histogram Components Touched By Developer - For components touched commit



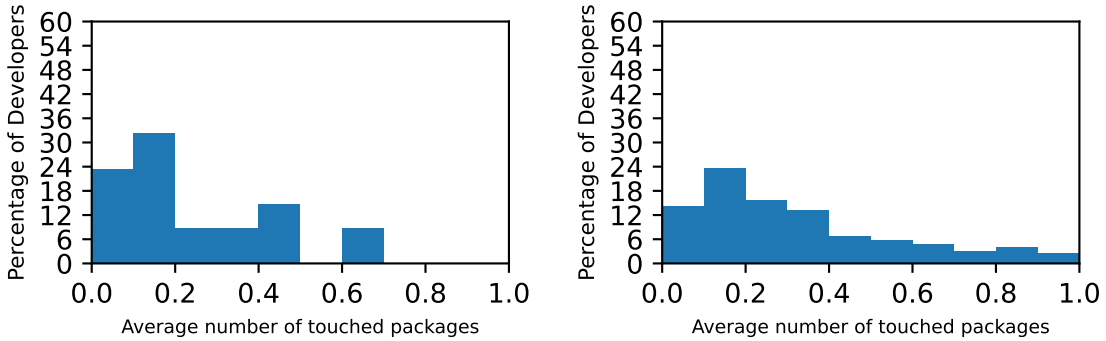
(a) Transient founder (n=193, μ=1.05, med=1.0, mod=1.0, max y=84.5, max x=6.0)

(b) Transient later joiner (n=15099, μ=1.15, med=1.0, mod=1.0, max y=82.6, max x=59.0)



(c) Moderate founder (n=42, μ=0.57, med=0.4, mod=0.1, max y=33.3, max x=3.9)

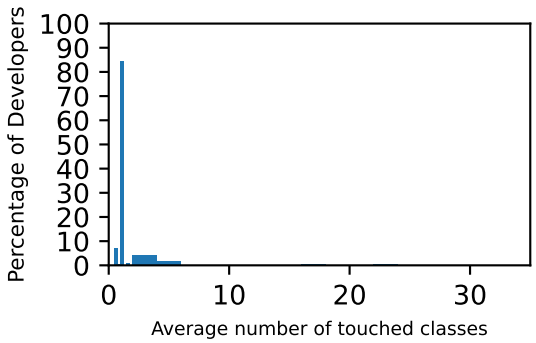
(d) Moderate later joiner (n=646, μ=0.79, med=0.7, mod=0.4, max y=23.4, max x=5.4)



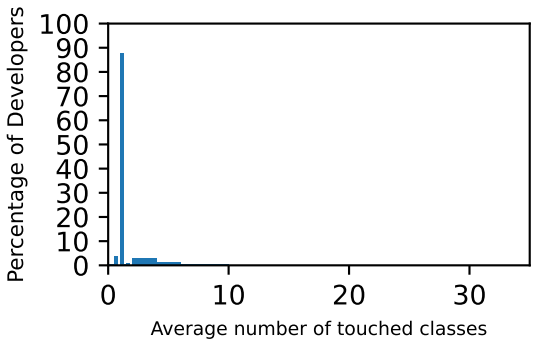
(e) Sustained founder (n=34, μ=0.23, med=0.1, mod=0.1, max y=32.4, max x=1.3)

(f) Sustained later joiner (n=203, μ=0.36, med=0.2, mod=0.1, max y=23.6, max x=4.0)

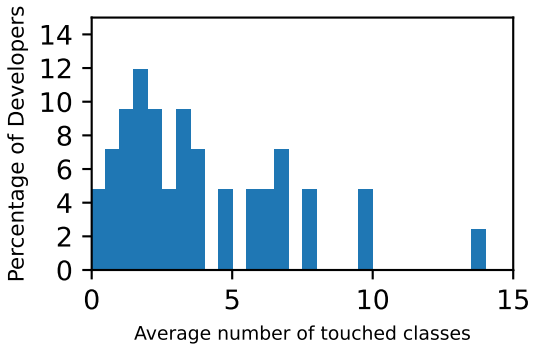
Figure 2: Histogram of percentage of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



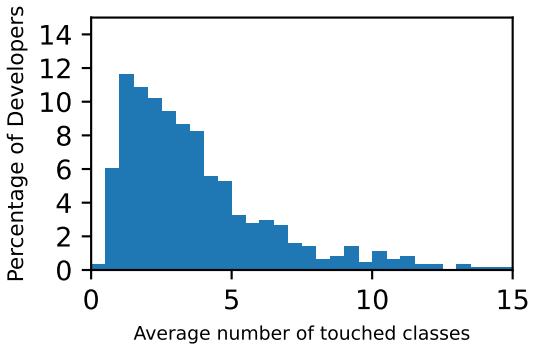
(a) Transient founder (n=193, μ=1.31, med=1.0, mod=1.0, max y=84.5, max x=23.5)



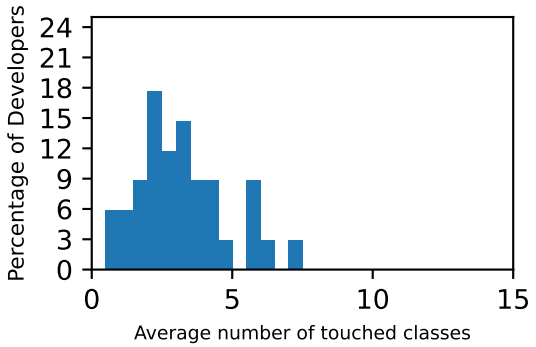
(b) Transient later joiner (n=15099, μ=1.3, med=1.0, mod=1.0, max y=87.6, max x=97.0)



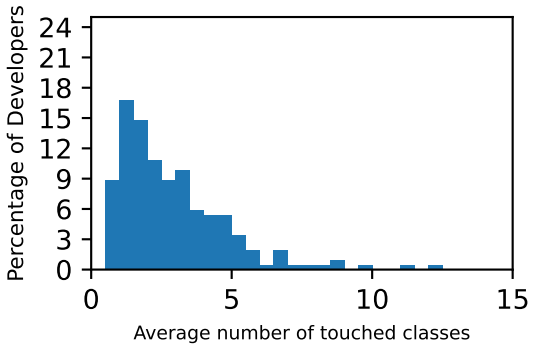
(c) Moderate founder (n=42, μ=4.58, med=3.0, mod=2.4, max y=11.9, max x=36.0)



(d) Moderate later joiner (n=646, μ=3.99, med=3.0, mod=1.1, max y=11.6, max x=38.6)

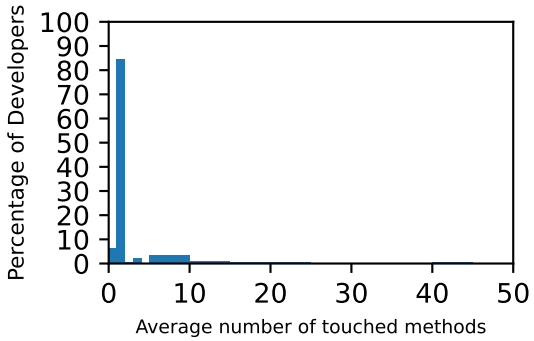


(e) Sustained founder (n=34, μ=3.14, med=2.95, mod=5.7, max y=17.6, max x=7.2)

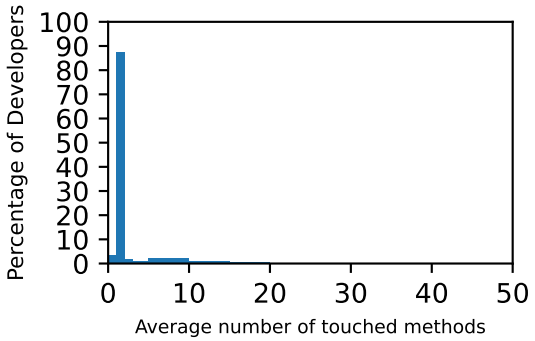


(f) Sustained later joiner (n=203, μ=3.19, med=2.4, mod=1.1, max y=16.7, max x=36.1)

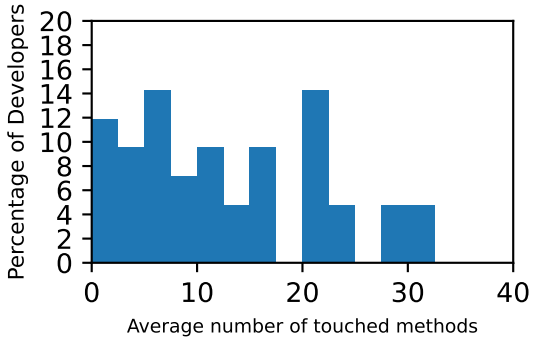
Figure 3: Histogram of percentage of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



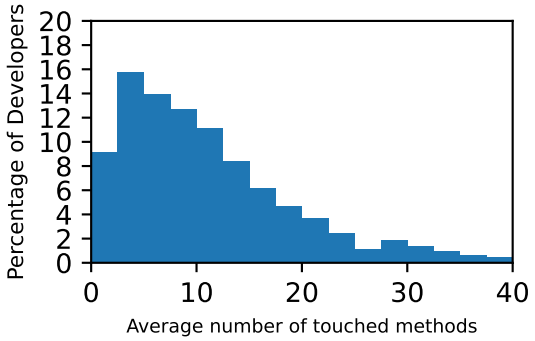
(a) Transient founder (n=193, μ=2.17, med=1.0, mod=1.0, max y=84.5, max x=76.0)



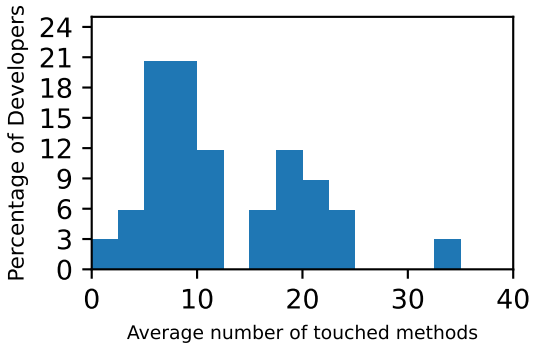
(b) Transient later joiner (n=15099, μ=2.25, med=1.0, mod=1.0, max y=87.5, max x=324.0)



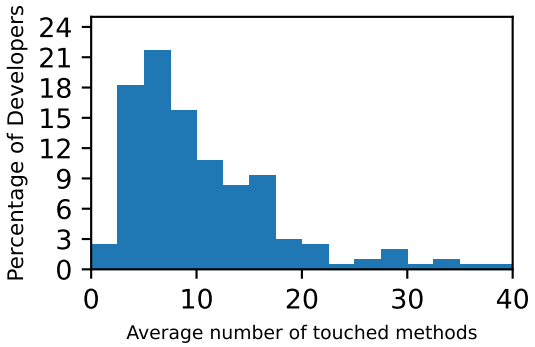
(c) Moderate founder (n=42, μ=20.11, med=11.5, mod=6.9, max y=14.3, max x=244.7)



(d) Moderate later joiner (n=646, μ=14.1, med=9.55, mod=6.2, max y=15.8, max x=155.9)



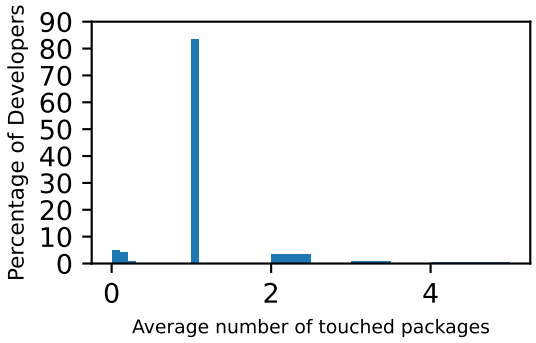
(e) Sustained founder (n=34, μ=13.28, med=10.15, mod=5.9, max y=20.6, max x=47.0)



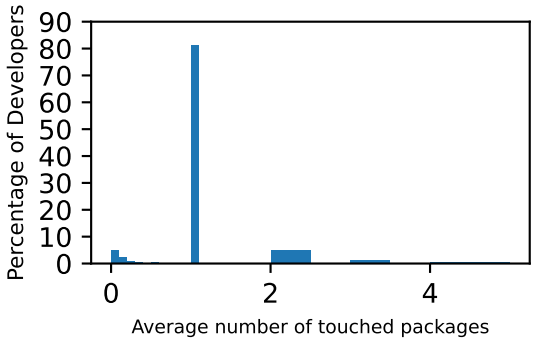
(f) Sustained later joiner (n=203, μ=12.67, med=8.8, mod=5.5, max y=21.7, max x=182.2)

Figure 4: Histogram of percentage of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.

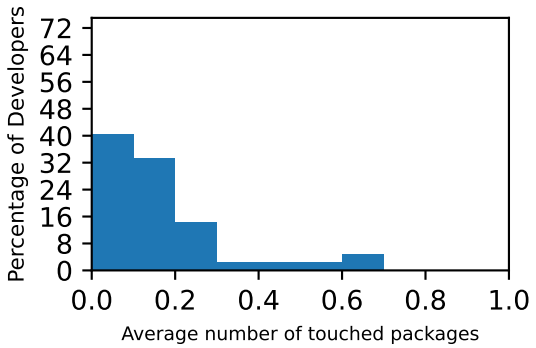
4.2 Histogram Components Touched By Developer - For components touched daily



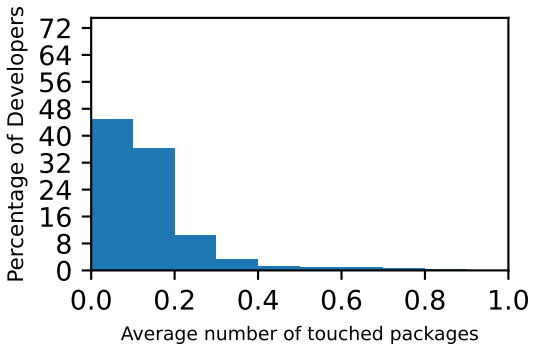
(a) Transient founder (n=193, μ=1.02, med=1.0, mod=1.0, max y=83.4, max x=6.0)



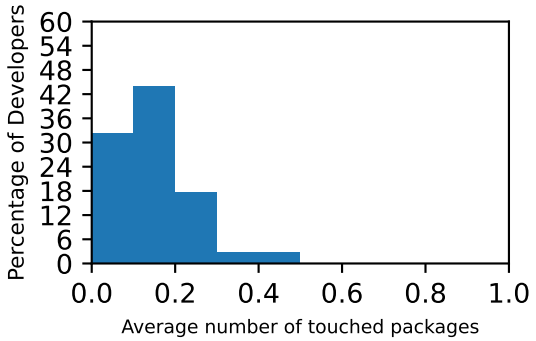
(b) Transient later joiner (n=15099, μ=1.08, med=1.0, mod=1.0, max y=81.3, max x=59.0)



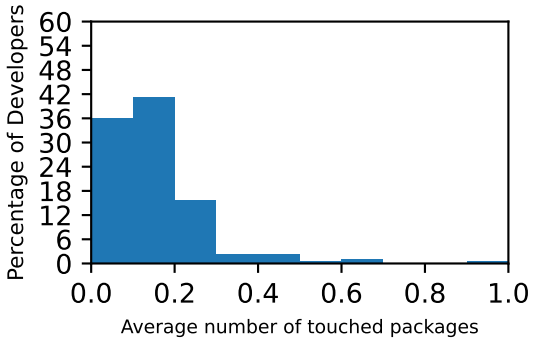
(c) Moderate founder (n=42, μ=0.12, med=0.1, mod=0.01, max y=40.5, max x=0.6)



(d) Moderate later joiner (n=646, μ=0.12, med=0.1, mod=0.01, max y=44.9, max x=11.0)

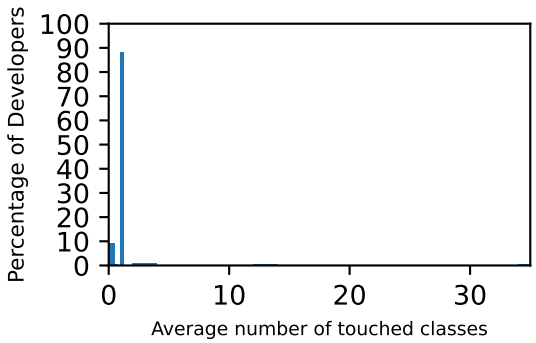


(e) Sustained founder (n=34, μ=0.1, med=0.1, mod=0.1, max y=44.1, max x=0.4)

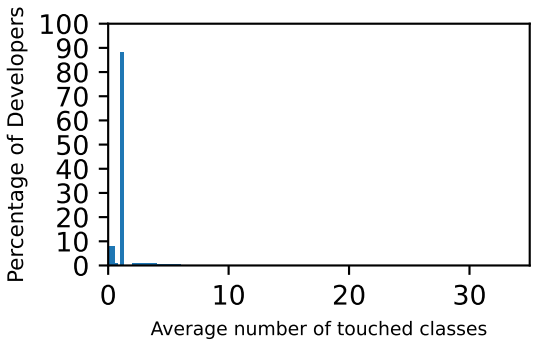


(f) Sustained later joiner (n=203, μ=0.11, med=0.1, mod=0.1, max y=41.4, max x=1.0)

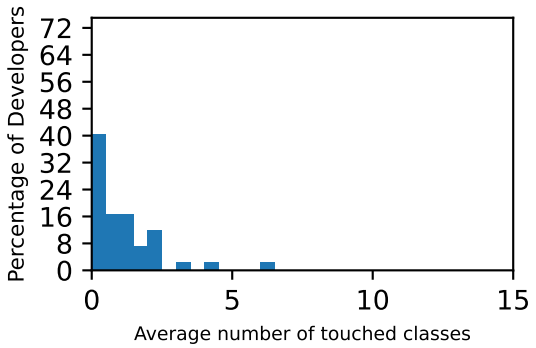
Figure 5: Histogram of percentage of developers (y-axis) against the average packages touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



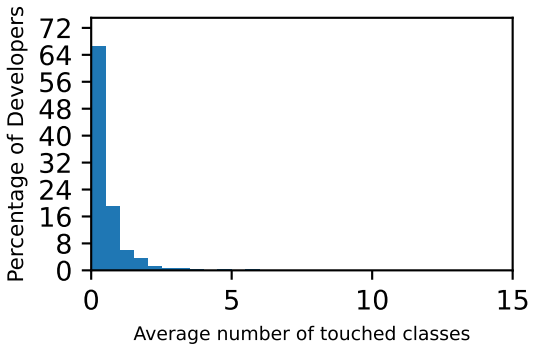
(a) Transient founder (n=193, μ=1.17, med=1.0, mod=1.0, max y=88.1, max x=35.0)



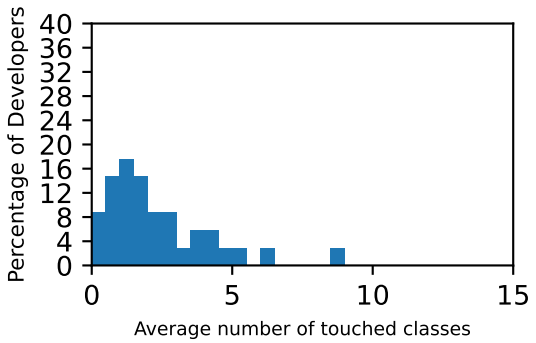
(b) Transient later joiner (n=15099, μ=1.05, med=1.0, mod=1.0, max y=88.1, max x=97.0)



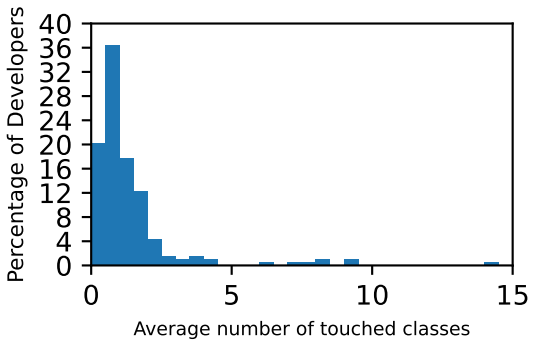
(c) Moderate founder (n=42, μ=1.07, med=0.7, mod=0.1, max y=40.5, max x=6.3)



(d) Moderate later joiner (n=646, μ=0.58, med=0.3, mod=0.1, max y=66.7, max x=15.0)

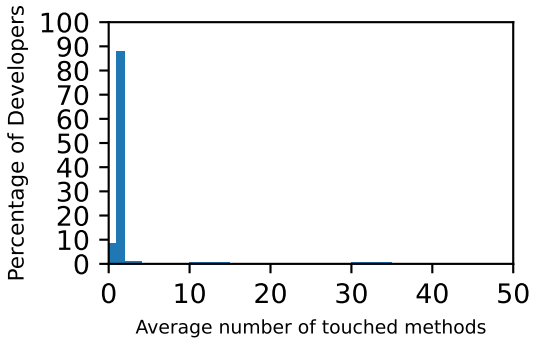


(e) Sustained founder (n=34, μ=2.31, med=1.9, mod=1.9, max y=17.6, max x=8.6)

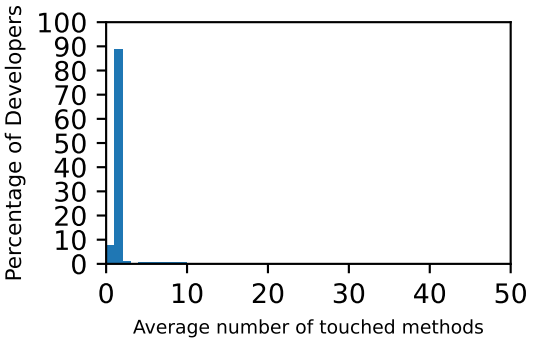


(f) Sustained later joiner (n=203, μ=1.32, med=0.9, mod=0.5, max y=36.5, max x=14.1)

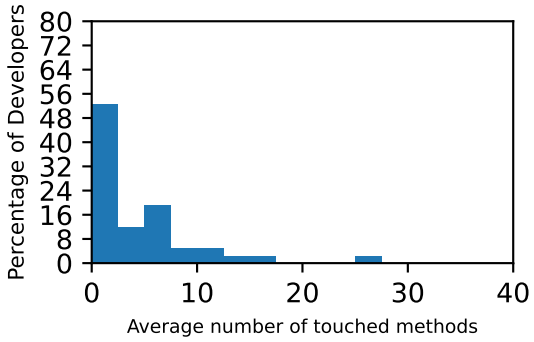
Figure 6: Histogram of percentage of developers (y-axis) against the average classes touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.



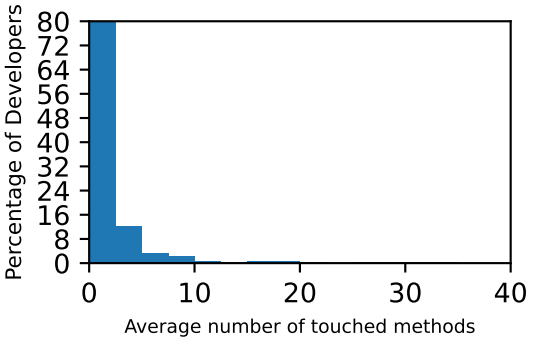
(a) Transient founder (n=193, μ=1.75, med=1.0, mod=1.0, max y=88.1, max x=109.0)



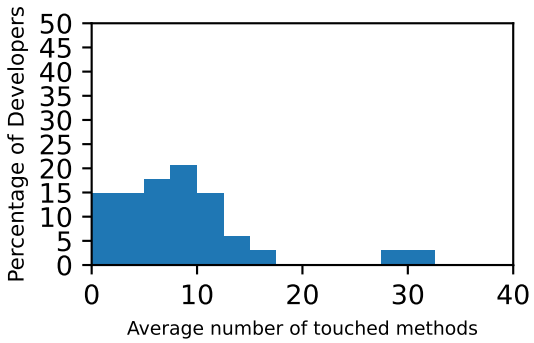
(b) Transient later joiner (n=15099, μ=1.38, med=1.0, mod=1.0, max y=88.7, max x=254.0)



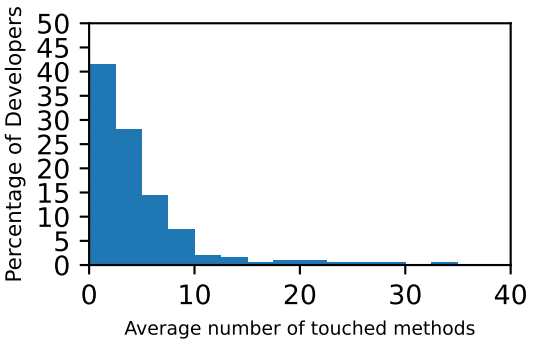
(c) Moderate founder (n=42, μ=4.34, med=2.35, mod=0.5, max y=52.4, max x=26.0)



(d) Moderate later joiner (n=646, μ=2.11, med=0.9, mod=0.3, max y=79.6, max x=76.3)



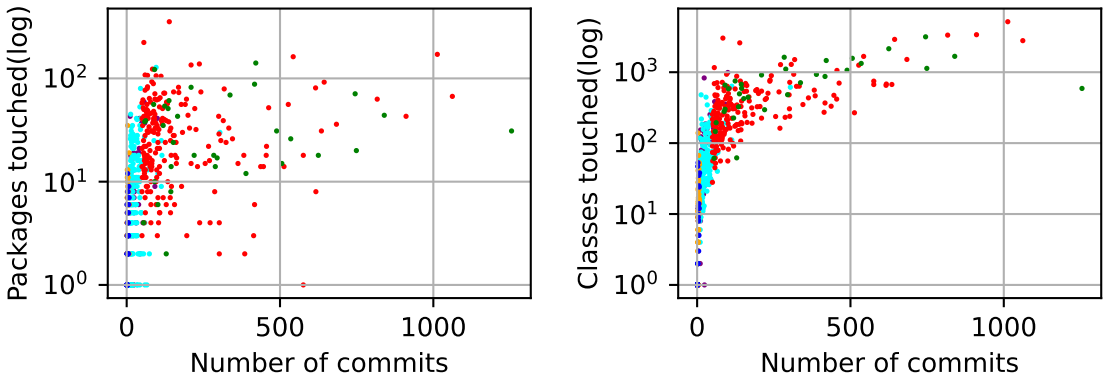
(e) Sustained founder (n=34, μ=9.61, med=7.7, mod=7.8, max y=20.6, max x=44.0)



(f) Sustained later joiner (n=203, μ=5.02, med=3.1, mod=1.0, max y=41.4, max x=53.0)

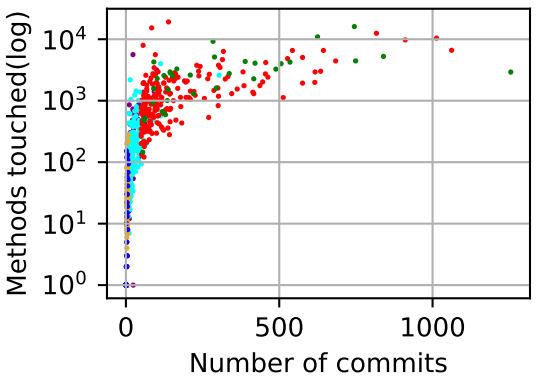
Figure 7: Histogram of percentage of developers (y-axis) against the average methods touched by daily (x-axis) from six (6) categories of developers from 31 repositories sampled from GitHub.

5 Sample Scatter Touched Against Commit -



(a) Packages

(b) Classes



(c) Methods

Figure 8: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at NUM4 sustained later joiners from six (6) categories: Transient Founder (Blue, 193), Moderate Founder (Purple, 42), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

6 Normality And Variance - For all components touched for all commits

Certain statistical methods require normality of each sample or the equality of variance. A Shapiro-Wilk Test evaluates whether a sample distribution significantly differs from the normal distribution. A Levene's test evaluates if the independent samples have equal variance.

Columns below:

- **Component** - Packages, classes or methods touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A.
- **Sample A Number** - The number of developers the first sample A.
- **Sample B** μ - The mean average of the second sample B.
- **Sample B Number** - The number of developers the first sample B.
- **Percentage Difference in** μ - The percentage difference in means between samples.
- **Sample A Normality** P - Shapiro normality P value of sample A.
- **Sample B Normality** P - Shapiro normality P value of sample B.
- **Variance Equality** P - The Levene variance equality p-value between sample A and B.
- **Statistical Model** - A recommended statistical model of Welch t test; Mann-Whitney; Brunner Münzel
- **Model** P - The P value of the recommended test Welch t test; Mann-Whitney; Brunner Münzel

Table 2: A Shapiro-Wilk test of packages for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Founder		Joiner		Percentage Difference in μ	Founder Normality P	Joiner Normality P	Variance Equality P	Statistical Models	Model P
			μ	number	μ	number						
Packages	1	All	1.54	269	1.22	15948	79.35	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Moderate	2.55	42	2.59	646	101.84	<0.001	<0.001	0.7955	Mann-Whitney	0.8829
Packages	1	Sustained	2.21	34	2.40	203	108.98	<0.001	<0.001	0.6501	Mann-Whitney	0.8805
Packages	2	All	3.08	110	2.97	2978	96.51	<0.001	<0.001	0.9917	Mann-Whitney	0.4243
Packages	2	Moderate	3.48	42	3.58	646	103.00	<0.001	<0.001	0.9633	Mann-Whitney	0.5758
Packages	2	Sustained	3.00	34	3.57	203	118.88	<0.001	<0.001	0.4130	Mann-Whitney	0.7985
Packages	5	Transient with five or more; moderate & sustained	4.98	85	6.09	1278	122.44	<0.001	<0.001	0.1874	Mann-Whitney	0.0425
Packages	5	Moderate	5.26	42	6.40	646	121.59	<0.001	<0.001	0.4213	Mann-Whitney	0.0674
Packages	5	Sustained	4.74	34	6.13	203	129.41	<0.001	<0.001	0.3795	Mann-Whitney	0.2268
Packages	10	Moderate & sustained	7.25	75	9.45	783	130.31	<0.001	<0.001	0.3686	Mann-Whitney	0.0010
Packages	10	Moderate	7.78	41	9.73	580	125.09	<0.001	<0.001	0.7655	Mann-Whitney	0.0116
Packages	10	Sustained	6.62	34	8.65	203	130.71	<0.001	<0.001	0.4753	Mann-Whitney	0.1076
Packages	20	Sustained	9.32	34	13.06	203	140.12	<0.001	<0.001	0.3136	Mann-Whitney	0.0435

Table 3: A Shapiro-Wilk test of packages for sample A category versus sample B category sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Sample				Percentage Difference in μ	Sample		Variance Equality P	Statistical Models	Model P
			A μ	A number	B μ	B number		A Normality P	B Normality P			
Packages	1	Transient vs. moderate	1.15	15292	2.59	688	225.27	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Transient vs. sustained	1.15	15292	2.38	237	206.49	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Transient vs. moderate & sustained	1.15	15292	2.54	925	220.46	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	1	Moderate vs. sustained	2.59	688	2.38	237	91.66	<0.001	<0.001	0.6005	Mann-Whitney	0.1045
Packages	1	Core vs. Non Core	1.87	198	1.34	16019	71.55	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Packages	2	Moderate vs. sustained	3.57	688	3.49	237	97.51	<0.001	<0.001	0.9982	Mann-Whitney	0.1265
Packages	2	Core vs. Non Core	3.72	112	2.95	2976	79.23	<0.001	<0.001	0.2942	Mann-Whitney	0.0000
Packages	5	Moderate vs. sustained	6.33	688	5.93	237	93.68	<0.001	<0.001	0.7055	Mann-Whitney	0.0805
Packages	5	Core vs. Non Core	6.21	107	6.01	1256	96.82	<0.001	<0.001	0.3528	Mann-Whitney	0.1946
Packages	10	Moderate vs. sustained	9.60	621	8.36	237	87.03	<0.001	<0.001	0.4692	Mann-Whitney	0.0006
Packages	10	Core vs. Non Core	9.12	104	9.28	754	101.68	<0.001	<0.001	0.2716	Mann-Whitney	0.8274
Packages	20	Core vs. Non Core	13.00	58	12.37	179	95.19	<0.001	<0.001	0.2640	Mann-Whitney	0.0461

Table 4: A Shapiro-Wilk test of classes for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Founder		Joiner		Percentage Difference in μ	Founder Normality P	Joiner Normality P	Variance Equality P	Statistical Models	Model P
			μ	number	μ	number						
Classes	1	All	2.83	269	1.27	15948	45.03	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Moderate	7.71	42	2.95	646	38.21	<0.001	<0.001	<0.001	Brunner-Münzel	0.4887
Classes	1	Sustained	3.44	34	5.49	203	159.47	<0.001	<0.001	0.7146	Mann-Whitney	0.1429
Classes	2	All	12.41	110	6.03	2978	48.60	<0.001	<0.001	<0.001	Brunner-Münzel	0.2607
Classes	2	Moderate	13.48	42	8.14	646	60.42	<0.001	<0.001	0.0402	Brunner-Münzel	0.3325
Classes	2	Sustained	19.62	34	8.25	203	42.04	<0.001	<0.001	0.1236	Mann-Whitney	0.0313
Classes	5	Transient with five or more; moderate & sustained	35.89	85	22.77	1278	63.43	<0.001	<0.001	0.0041	Brunner-Münzel	0.0800
Classes	5	Moderate	39.76	42	22.96	646	57.75	<0.001	<0.001	<0.001	Brunner-Münzel	0.2085
Classes	5	Sustained	36.32	34	30.01	203	82.62	<0.001	<0.001	0.6508	Mann-Whitney	0.5895
Classes	10	Moderate & sustained	63.96	75	43.51	783	68.03	<0.001	<0.001	<0.001	Brunner-Münzel	0.1269
Classes	10	Moderate	60.12	41	41.95	580	69.77	<0.001	<0.001	0.0043	Brunner-Münzel	0.2654
Classes	10	Sustained	68.59	34	47.98	203	69.95	<0.001	<0.001	0.1446	Mann-Whitney	0.2478
Classes	20	Sustained	107.50	34	81.91	203	76.20	<0.001	<0.001	0.0906	Mann-Whitney	0.2857

Table 5: A Shapiro-Wilk test of classes for sample A category versus sample B category sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Sample				Percentage Difference in μ	Sample		Variance Equality P	Statistical Models	Model P
			A μ	A number	B μ	B number		A Normality P	B Normality P			
Classes	1	Transient vs. moderate	1.15	15292	3.24	688	281.11	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Transient vs. sustained	1.15	15292	5.19	237	450.89	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Transient vs. moderate & sustained	1.15	15292	3.74	925	324.61	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Classes	1	Moderate vs. sustained	3.24	688	5.19	237	160.39	<0.001	<0.001	0.1246	Mann-Whitney	0.2623
Classes	1	Core vs. Non Core	3.67	198	1.28	16019	34.99	<0.001	<0.001	<0.001	Brunner-Münzel	0.0001
Classes	2	Moderate vs. sustained	8.47	688	9.88	237	116.65	<0.001	<0.001	0.3965	Mann-Whitney	0.3370
Classes	2	Core vs. Non Core	12.88	112	6.01	2976	46.64	<0.001	<0.001	<0.001	Brunner-Münzel	0.0105
Classes	5	Moderate vs. sustained	23.99	688	30.92	237	128.88	<0.001	<0.001	0.0208	Brunner-Münzel	0.6889
Classes	5	Core vs. Non Core	30.73	107	22.98	1256	74.78	<0.001	<0.001	0.0219	Brunner-Münzel	0.7285
Classes	10	Moderate vs. sustained	43.15	621	50.94	237	118.05	<0.001	<0.001	0.0197	Brunner-Münzel	0.6812
Classes	10	Core vs. Non Core	54.99	104	43.96	754	79.95	<0.001	<0.001	0.0157	Brunner-Münzel	0.9861
Classes	20	Core vs. Non Core	104.21	58	79.55	179	76.34	<0.001	<0.001	0.5019	Mann-Whitney	0.0014

Table 6: A Shapiro-Wilk test of methods for different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Founder		Joiner		Percentage Difference in μ	Founder Normality P	Joiner Normality P	Variance Equality P	Statistical Models	Model P
			μ	number	μ	number						
Methods	1	All	7.16	269	2.01	15948	28.08	<0.001	<0.001	0.0037	Brunner-Münzel	0.0000
Methods	1	Moderate	23.71	42	7.92	646	33.42	<0.001	<0.001	0.0033	Brunner-Münzel	0.5660
Methods	1	Sustained	10.59	34	24.08	203	227.41	<0.001	<0.001	0.7393	Mann-Whitney	0.2351
Methods	2	All	54.12	110	19.93	2978	36.84	<0.001	<0.001	<0.001	Brunner-Münzel	0.4015
Methods	2	Moderate	61.76	42	29.98	646	48.54	<0.001	<0.001	0.0155	Brunner-Münzel	0.4320
Methods	2	Sustained	88.32	34	25.84	203	29.26	<0.001	<0.001	0.0456	Brunner-Münzel	0.0667
Methods	5	Transient with five or more; moderate & sustained	149.40	85	80.32	1278	53.76	<0.001	<0.001	0.0026	Brunner-Münzel	0.0836
Methods	5	Moderate	179.29	42	80.93	646	45.14	<0.001	<0.001	<0.001	Brunner-Münzel	0.1635
Methods	5	Sustained	138.53	34	114.77	203	82.85	<0.001	<0.001	0.7156	Mann-Whitney	0.7890
Methods	10	Moderate & sustained	259.76	75	154.41	783	59.44	<0.001	<0.001	0.0019	Brunner-Münzel	0.0920
Methods	10	Moderate	264.83	41	145.21	580	54.83	<0.001	<0.001	0.0018	Brunner-Münzel	0.2351
Methods	10	Sustained	253.65	34	180.70	203	71.24	<0.001	<0.001	0.3267	Mann-Whitney	0.2291
Methods	20	Sustained	401.24	34	295.10	203	73.55	<0.001	<0.001	0.1120	Mann-Whitney	0.5131

Table 7: A Shapiro-Wilk test of methods for sample A category versus sample B category sample normality and a Levene's variance test between samples

Component	First n commits	Contributory Category	Sample				Percentage Difference in μ	Sample		Variance Equality P	Statistical Models	Model P
			A μ	A number	B μ	B number		A Normality P	B Normality P			
Methods	1	Transient vs. moderate	1.48	15292	8.89	688	600.13	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Transient vs. sustained	1.48	15292	22.14	237	1495.13	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Transient vs. moderate & sustained	1.48	15292	12.28	925	829.44	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	1	Moderate vs. sustained	8.89	688	22.14	237	249.14	<0.001	<0.001	0.1231	Mann-Whitney	0.1376
Methods	1	Core vs. Non Core	9.51	198	2.05	16019	21.56	<0.001	<0.001	<0.001	Brunner-Münzel	0.0000
Methods	2	Moderate vs. sustained	31.92	688	34.81	237	109.05	<0.001	<0.001	0.7249	Mann-Whitney	0.4271
Methods	2	Core vs. Non Core	52.45	112	19.97	2976	38.09	<0.001	<0.001	<0.001	Brunner-Münzel	0.0032
Methods	5	Moderate vs. sustained	86.94	688	118.18	237	135.93	<0.001	<0.001	0.0621	Mann-Whitney	0.8232
Methods	5	Core vs. Non Core	119.94	107	81.62	1256	68.05	<0.001	<0.001	0.0336	Brunner-Münzel	0.4154
Methods	10	Moderate vs. sustained	153.10	621	191.16	237	124.86	<0.001	<0.001	0.0473	Brunner-Münzel	0.9101
Methods	10	Core vs. Non Core	210.71	104	157.12	754	74.57	<0.001	<0.001	0.0194	Brunner-Münzel	0.3184
Methods	20	Core vs. Non Core	347.88	58	298.16	179	85.71	<0.001	<0.001	0.6980	Mann-Whitney	0.1034

7 ART Anova - For all components touched for all commits

An **Aligned Rank Transform (ART)** Anova table for all components touched on average. The statistical significance of architectural exploration is evaluated using a dual-framework modeling methodology. A standard parametric ordinary least squares (**OLS**) ANOVA is presented alongside a robust non-parametric ANOVA. This side-by-side presentation explicitly demonstrates where extreme right-skew and outlier-induced variance in repository mining data mask true behavioral effects in traditional linear models, which the rank-transformed framework uncovers.

Explanation of Combined Table Columns:

- **Component / Commits (n) / Devs (N):** Define the experimental scope, tracking the architectural tier (packages, classes, or methods), the longitudinal lifecycle snapshot threshold, and the active developer population size.
- **Factor:** The independent categorical variables or interactions being tested for statistical significance.
- **OLS df:** Parametric numerator degrees of freedom based on factor grouping configurations.
- **OLS F-Stat:** The traditional variance ratio metric calculated from raw values.
- **OLS P-Value:** Parametric significance probability. Note how heteroscedastic noise frequently inflates these values toward non-significance ($p > 0.05$).
- **OLS η^2 (Eta-Squared):** The total proportion of raw variance explained exclusively by this standalone factor.
- **ART df (Resid):** Formatted as df_{effect} (df_{residual}). Because the ART methodology conducts independent alignment procedures for each factor before ranking, it assigns the exact denominator error degrees of freedom dynamically alongside its numerator.
- **ART F-Stat:** The robust variance ratio calculated over the aligned, ranked data layers.
- **ART P-Value:** Non-parametric significance probability, providing the authoritative test for heavy-tailed software metrics.
- **ART Part. η^2 (Partial Eta-Squared):** The effect size over ranked data layers, indicating the proportion of variance explained by this specific factor after controlling for alternate main effects.

Explanation of Fixed Table Rows:

- **Role (F/LJ):** Isolates the *Main Role Effect* to evaluate whether the timing of project entry (Founder vs. Later Joiner) significantly alters exploration breadth on its own.
- **Category (T/M/S or M/S):** Isolates the *Main Category Effect* to determine whether a contributor's eventual longevity footprint (Transient, Moderate, or Sustained) dictates early behavior.
- **Interaction:** Evaluates the *Interaction Effect* (Role $\times$ Category). A significant interaction indicates that the impact of entry role is not uniform, but behaves completely differently depending on the eventual tenure category of the contributor.

Table 8: ART Anova results for all components, number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	OLS				ART			
				df	F-Stat	P Value	η^2	df (Resid)	F-Stat	P Value	Part. η^2
packages	1	16217	Role (F/LJ)	1	0.07	0.7976	0.0000	1 (16211)	259.78	0.0000	0.0158
packages	1	16217	Category (T/M/S)	2	30.11	0.0000	0.0037	2 (16211)	774.10	0.0000	0.0872
packages	1	16217	Interaction	2	0.69	0.5007	0.0001	2 (16211)	123.16	0.0000	0.0150
packages	2	3088	Role (F/LJ)	1	0.01	0.9062	0.0000	1 (3082)	5.61	0.0180	0.0018
packages	2	3088	Category (M/S)	2	0.27	0.7664	0.0002	2 (3082)	8.02	0.0003	0.0052
packages	2	3088	Interaction	2	0.12	0.8843	0.0001	2 (3082)	0.33	0.7217	0.0002
packages	5	1363	Role (F/LJ)	1	0.01	0.9104	0.0000	1 (1357)	0.13	0.7154	0.0001
packages	5	1363	Category (M/S)	2	0.16	0.8494	0.0002	2 (1357)	4.49	0.0114	0.0066
packages	5	1363	Interaction	2	0.11	0.8936	0.0002	2 (1357)	0.06	0.9412	0.0001
packages	10	858	Role (F/LJ)	1	0.01	0.9350	0.0000	1 (854)	0.41	0.5210	0.0005
packages	10	858	Category (M/S)	1	0.34	0.5611	0.0004	1 (854)	0.64	0.4256	0.0007
packages	10	858	Interaction	1	0.12	0.7340	0.0001	1 (854)	1.11	0.2920	0.0013
classes	1	16217	Role (F/LJ)	1	45.17	0.0000	0.0027	1 (16211)	457.31	0.0000	0.0274
classes	1	16217	Category (T/M/S)	2	32.25	0.0000	0.0039	2 (16211)	870.90	0.0000	0.0970
classes	1	16217	Interaction	2	21.95	0.0000	0.0027	2 (16211)	289.32	0.0000	0.0345
classes	2	3088	Role (F/LJ)	1	8.65	0.0033	0.0028	1 (3082)	216.13	0.0000	0.0655
classes	2	3088	Category (M/S)	2	1.78	0.1680	0.0011	2 (3082)	24.81	0.0000	0.0158
classes	2	3088	Interaction	2	3.83	0.0219	0.0024	2 (3082)	9.42	0.0001	0.0061
classes	5	1363	Role (F/LJ)	1	4.55	0.0330	0.0033	1 (1357)	17.59	0.0000	0.0128
classes	5	1363	Category (M/S)	2	0.88	0.4137	0.0013	2 (1357)	18.41	0.0000	0.0264
classes	5	1363	Interaction	2	2.13	0.1198	0.0031	2 (1357)	4.22	0.0149	0.0062
classes	10	858	Role (F/LJ)	1	3.01	0.0832	0.0035	1 (854)	24.09	0.0000	0.0274
classes	10	858	Category (M/S)	1	1.20	0.2745	0.0014	1 (854)	52.66	0.0000	0.0581
classes	10	858	Interaction	1	2.61	0.1065	0.0030	1 (854)	12.28	0.0005	0.0142
methods	1	16217	Role (F/LJ)	1	11.96	0.0005	0.0007	1 (16211)	376.15	0.0000	0.0227
methods	1	16217	Category (T/M/S)	2	9.31	0.0001	0.0011	2 (16211)	767.04	0.0000	0.0865
methods	1	16217	Interaction	2	8.84	0.0001	0.0011	2 (16211)	375.25	0.0000	0.0442
methods	2	3088	Role (F/LJ)	1	2.28	0.1314	0.0007	1 (3082)	254.68	0.0000	0.0763
methods	2	3088	Category (M/S)	2	0.46	0.6288	0.0003	2 (3082)	38.77	0.0000	0.0245
methods	2	3088	Interaction	2	1.72	0.1800	0.0011	2 (3082)	24.78	0.0000	0.0158
methods	5	1363	Role (F/LJ)	1	1.11	0.2925	0.0008	1 (1357)	40.73	0.0000	0.0291
methods	5	1363	Category (M/S)	2	0.18	0.8314	0.0003	2 (1357)	27.48	0.0000	0.0389
methods	5	1363	Interaction	2	0.84	0.4301	0.0012	2 (1357)	7.01	0.0009	0.0102
methods	10	858	Role (F/LJ)	1	0.74	0.3897	0.0009	1 (854)	78.44	0.0000	0.0841
methods	10	858	Category (M/S)	1	0.25	0.6180	0.0003	1 (854)	66.08	0.0000	0.0718
methods	10	858	Interaction	1	1.06	0.3034	0.0012	1 (854)	11.37	0.0008	0.0131

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Later Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{2}$$

8 Holm-Bonferroni—For all components touched for all commits

A Pairwise Post-hoc Analysis with Holm-Bonferroni Correction table for all components touched on average.

Post-hoc Pairwise Analysis (Holm-Bonferroni): Following the multi-factor significance analysis, non-parametric pairwise comparisons were executed to isolate directional stochastic dominance patterns between groups. To counteract family-wise Type I error inflation across multiple group permutations without experiencing the severe statistical power depletion of a traditional Bonferroni adjustment, a sequentially rejective Holm-Bonferroni step-down correction is applied.

Table 9: Significant Holm-Bonferroni Pairwise results for packages and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
packages	1	16217	Founder Moderate	Founder Transient	1.34	0.0000
packages	1	16217	Founder Moderate	Later Joiner Transient	1.40	0.0000
packages	1	16217	Founder Sustained	Founder Transient	1.00	0.0000
packages	1	16217	Founder Sustained	Later Joiner Transient	1.06	0.0000
packages	1	16217	Founder Transient	Later Joiner Moderate	-1.39	0.0000
packages	1	16217	Founder Transient	Later Joiner Sustained	-1.20	0.0000
packages	1	16217	Later Joiner Moderate	Later Joiner Transient	1.44	0.0000
packages	1	16217	Later Joiner Sustained	Later Joiner Transient	1.25	0.0000
packages	2	3088	Later Joiner Moderate	Later Joiner Transient	0.53	0.0000
packages	5	1363	Later Joiner Moderate	Later Joiner Transient	0.53	0.0214

Table 10: Significant Holm-Bonferroni Pairwise results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16217	Founder Moderate	Founder Transient	6.06	0.0004
classes	1	16217	Founder Moderate	Later Joiner Transient	6.57	0.0000
classes	1	16217	Founder Sustained	Founder Transient	1.78	0.0002
classes	1	16217	Founder Sustained	Later Joiner Transient	2.30	0.0000
classes	1	16217	Founder Transient	Later Joiner Moderate	-1.29	0.0000
classes	1	16217	Founder Transient	Later Joiner Sustained	-3.83	0.0017
classes	1	16217	Founder Transient	Later Joiner Transient	0.51	0.0000
classes	1	16217	Later Joiner Moderate	Later Joiner Transient	1.80	0.0000
classes	1	16217	Later Joiner Sustained	Later Joiner Transient	4.34	0.0000
classes	2	3088	Founder Transient	Later Joiner Sustained	-0.75	0.0198
classes	2	3088	Founder Transient	Later Joiner Transient	2.70	0.0002
classes	2	3088	Later Joiner Moderate	Later Joiner Transient	0.92	0.0001
classes	5	1363	Later Joiner Moderate	Later Joiner Transient	1.01	0.0340

Table 11: Significant Holm-Bonferroni Pairwise results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16217	Founder Moderate	Founder Transient	20.76	0.0004
methods	1	16217	Founder Moderate	Later Joiner Transient	22.25	0.0000
methods	1	16217	Founder Sustained	Founder Transient	7.63	0.0007
methods	1	16217	Founder Sustained	Later Joiner Transient	9.13	0.0000
methods	1	16217	Founder Transient	Later Joiner Moderate	-4.97	0.0000
methods	1	16217	Founder Transient	Later Joiner Sustained	-21.12	0.0010
methods	1	16217	Founder Transient	Later Joiner Transient	1.50	0.0000
methods	1	16217	Later Joiner Moderate	Later Joiner Transient	6.46	0.0000
methods	1	16217	Later Joiner Sustained	Later Joiner Transient	22.62	0.0000
methods	2	3088	Founder Transient	Later Joiner Sustained	-11.96	0.0052
methods	2	3088	Founder Transient	Later Joiner Transient	7.84	0.0000
methods	2	3088	Later Joiner Moderate	Later Joiner Transient	3.65	0.0000
methods	5	1363	Later Joiner Moderate	Later Joiner Transient	4.43	0.0086

Table 12: Not Significant one and two commits Holm-Bonferroni Pairwise results for packages and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
packages	1	16217	Founder Moderate	Founder Sustained	0.34	1.0000
packages	1	16217	Founder Moderate	Later Joiner Moderate	-0.05	1.0000
packages	1	16217	Founder Moderate	Later Joiner Sustained	0.14	1.0000
packages	1	16217	Founder Sustained	Later Joiner Moderate	-0.39	1.0000
packages	1	16217	Founder Sustained	Later Joiner Sustained	-0.20	1.0000
packages	1	16217	Founder Transient	Later Joiner Transient	0.06	0.9459
packages	1	16217	Later Joiner Moderate	Later Joiner Sustained	0.19	0.9459
packages	2	3088	Founder Moderate	Founder Sustained	0.34	1.0000
packages	2	3088	Founder Moderate	Founder Transient	0.37	1.0000
packages	2	3088	Founder Moderate	Later Joiner Moderate	-0.05	1.0000
packages	2	3088	Founder Moderate	Later Joiner Sustained	0.14	1.0000
packages	2	3088	Founder Moderate	Later Joiner Transient	0.49	0.6570
packages	2	3088	Founder Sustained	Founder Transient	0.03	1.0000
packages	2	3088	Founder Sustained	Later Joiner Moderate	-0.39	1.0000
packages	2	3088	Founder Sustained	Later Joiner Sustained	-0.20	1.0000
packages	2	3088	Founder Sustained	Later Joiner Transient	0.14	1.0000
packages	2	3088	Founder Transient	Later Joiner Moderate	-0.42	1.0000
packages	2	3088	Founder Transient	Later Joiner Sustained	-0.23	1.0000
packages	2	3088	Founder Transient	Later Joiner Transient	0.11	1.0000
packages	2	3088	Later Joiner Moderate	Later Joiner Sustained	0.19	1.0000
packages	2	3088	Later Joiner Sustained	Later Joiner Transient	0.34	0.6509

Table 13: Not Significant five and ten comits Holm-Bonferroni Pairwise results for packages and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
packages	5	1363	Founder Moderate	Founder Sustained	0.34	1.0000
packages	5	1363	Founder Moderate	Founder Transient	0.21	1.0000
packages	5	1363	Founder Moderate	Later Joiner Moderate	-0.05	1.0000
packages	5	1363	Founder Moderate	Later Joiner Sustained	0.14	1.0000
packages	5	1363	Founder Moderate	Later Joiner Transient	0.49	1.0000
packages	5	1363	Founder Sustained	Founder Transient	-0.13	1.0000
packages	5	1363	Founder Sustained	Later Joiner Moderate	-0.39	1.0000
packages	5	1363	Founder Sustained	Later Joiner Sustained	-0.20	1.0000
packages	5	1363	Founder Sustained	Later Joiner Transient	0.15	1.0000
packages	5	1363	Founder Transient	Later Joiner Moderate	-0.26	1.0000
packages	5	1363	Founder Transient	Later Joiner Sustained	-0.07	1.0000
packages	5	1363	Founder Transient	Later Joiner Transient	0.27	1.0000
packages	5	1363	Later Joiner Moderate	Later Joiner Sustained	0.19	1.0000
packages	5	1363	Later Joiner Sustained	Later Joiner Transient	0.34	1.0000
packages	10	858	Founder Moderate	Founder Sustained	0.38	1.0000
packages	10	858	Founder Moderate	Later Joiner Moderate	0.04	1.0000
packages	10	858	Founder Moderate	Later Joiner Sustained	0.18	1.0000
packages	10	858	Founder Sustained	Later Joiner Moderate	-0.34	1.0000
packages	10	858	Founder Sustained	Later Joiner Sustained	-0.20	1.0000
packages	10	858	Later Joiner Moderate	Later Joiner Sustained	0.14	0.8962
packages	20	237	Founder Sustained	Later Joiner Sustained	-0.20	0.8805

Table 14: Not Significant one and two commits Holm-Bonferroni Pairwise results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16217	Founder Moderate	Founder Sustained	4.27	1.0000
classes	1	16217	Founder Moderate	Later Joiner Moderate	4.77	1.0000
classes	1	16217	Founder Moderate	Later Joiner Sustained	2.23	0.8575
classes	1	16217	Founder Sustained	Later Joiner Moderate	0.49	1.0000
classes	1	16217	Founder Sustained	Later Joiner Sustained	-2.05	0.8575
classes	1	16217	Later Joiner Moderate	Later Joiner Sustained	-2.54	0.8575
classes	2	3088	Founder Moderate	Founder Sustained	4.27	1.0000
classes	2	3088	Founder Moderate	Founder Transient	2.98	1.0000
classes	2	3088	Founder Moderate	Later Joiner Moderate	4.77	1.0000
classes	2	3088	Founder Moderate	Later Joiner Sustained	2.23	1.0000
classes	2	3088	Founder Moderate	Later Joiner Transient	5.68	0.2788
classes	2	3088	Founder Sustained	Founder Transient	-1.29	1.0000
classes	2	3088	Founder Sustained	Later Joiner Moderate	0.49	1.0000
classes	2	3088	Founder Sustained	Later Joiner Sustained	-2.05	1.0000
classes	2	3088	Founder Sustained	Later Joiner Transient	1.41	0.2530
classes	2	3088	Founder Transient	Later Joiner Moderate	1.79	0.1124
classes	2	3088	Later Joiner Moderate	Later Joiner Sustained	-2.54	1.0000
classes	2	3088	Later Joiner Sustained	Later Joiner Transient	3.46	1.0000

Table 15: Not Significant five and ten comits Holm-Bonferroni Pairwise results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	5	1363	Founder Moderate	Founder Sustained	4.27	1.0000
classes	5	1363	Founder Moderate	Founder Transient	1.16	1.0000
classes	5	1363	Founder Moderate	Later Joiner Moderate	4.77	1.0000
classes	5	1363	Founder Moderate	Later Joiner Sustained	2.23	1.0000
classes	5	1363	Founder Moderate	Later Joiner Transient	5.78	0.5079
classes	5	1363	Founder Sustained	Founder Transient	-3.11	1.0000
classes	5	1363	Founder Sustained	Later Joiner Moderate	0.49	1.0000
classes	5	1363	Founder Sustained	Later Joiner Sustained	-2.05	1.0000
classes	5	1363	Founder Sustained	Later Joiner Transient	1.51	0.4285
classes	5	1363	Founder Transient	Later Joiner Moderate	3.61	1.0000
classes	5	1363	Founder Transient	Later Joiner Sustained	1.07	0.6014
classes	5	1363	Founder Transient	Later Joiner Transient	4.62	0.3126
classes	5	1363	Later Joiner Moderate	Later Joiner Sustained	-2.54	1.0000
classes	5	1363	Later Joiner Sustained	Later Joiner Transient	3.55	1.0000
classes	10	858	Founder Moderate	Founder Sustained	4.44	1.0000
classes	10	858	Founder Moderate	Later Joiner Moderate	4.90	1.0000
classes	10	858	Founder Moderate	Later Joiner Sustained	2.39	0.7561
classes	10	858	Founder Sustained	Later Joiner Moderate	0.47	1.0000
classes	10	858	Founder Sustained	Later Joiner Sustained	-2.05	0.7561
classes	10	858	Later Joiner Moderate	Later Joiner Sustained	-2.51	0.7561
classes	20	237	Founder Sustained	Later Joiner Sustained	-2.05	0.1429

Table 16: Not Significant one and two commits Holm-Bonferroni Pairwise results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16217	Founder Moderate	Founder Sustained	13.13	1.0000
methods	1	16217	Founder Moderate	Later Joiner Moderate	15.79	1.0000
methods	1	16217	Founder Moderate	Later Joiner Sustained	-0.36	0.8079
methods	1	16217	Founder Sustained	Later Joiner Moderate	2.66	1.0000
methods	1	16217	Founder Sustained	Later Joiner Sustained	-13.49	0.9405
methods	1	16217	Later Joiner Moderate	Later Joiner Sustained	-16.15	0.5367
methods	2	3088	Founder Moderate	Founder Sustained	13.13	1.0000
methods	2	3088	Founder Moderate	Founder Transient	11.60	1.0000
methods	2	3088	Founder Moderate	Later Joiner Moderate	15.79	1.0000
methods	2	3088	Founder Moderate	Later Joiner Sustained	-0.36	1.0000
methods	2	3088	Founder Moderate	Later Joiner Transient	19.44	0.3300
methods	2	3088	Founder Sustained	Founder Transient	-1.53	0.8051
methods	2	3088	Founder Sustained	Later Joiner Moderate	2.66	1.0000
methods	2	3088	Founder Sustained	Later Joiner Sustained	-13.49	1.0000
methods	2	3088	Founder Sustained	Later Joiner Transient	6.31	0.6086
methods	2	3088	Founder Transient	Later Joiner Moderate	4.19	0.0534
methods	2	3088	Later Joiner Moderate	Later Joiner Sustained	-16.15	0.8051
methods	2	3088	Later Joiner Sustained	Later Joiner Transient	19.80	1.0000

Table 17: Not Significant five and ten comits Holm-Bonferroni Pairwise results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	5	1363	Founder Moderate	Founder Sustained	13.13	1.0000
methods	5	1363	Founder Moderate	Founder Transient	8.05	1.0000
methods	5	1363	Founder Moderate	Later Joiner Moderate	15.79	1.0000
methods	5	1363	Founder Moderate	Later Joiner Sustained	-0.36	1.0000
methods	5	1363	Founder Moderate	Later Joiner Transient	20.22	0.5496
methods	5	1363	Founder Sustained	Founder Transient	-5.08	1.0000
methods	5	1363	Founder Sustained	Later Joiner Moderate	2.66	1.0000
methods	5	1363	Founder Sustained	Later Joiner Sustained	-13.49	1.0000
methods	5	1363	Founder Sustained	Later Joiner Transient	7.10	0.8187
methods	5	1363	Founder Transient	Later Joiner Moderate	7.74	0.6058
methods	5	1363	Founder Transient	Later Joiner Sustained	-8.41	0.2437
methods	5	1363	Founder Transient	Later Joiner Transient	12.17	0.0741
methods	5	1363	Later Joiner Moderate	Later Joiner Sustained	-16.15	0.8187
methods	5	1363	Later Joiner Sustained	Later Joiner Transient	20.59	1.0000
methods	10	858	Founder Moderate	Founder Sustained	13.68	1.0000
methods	10	858	Founder Moderate	Later Joiner Moderate	16.44	1.0000
methods	10	858	Founder Moderate	Later Joiner Sustained	0.19	0.6748
methods	10	858	Founder Sustained	Later Joiner Moderate	2.76	1.0000
methods	10	858	Founder Sustained	Later Joiner Sustained	-13.49	0.9405
methods	10	858	Later Joiner Moderate	Later Joiner Sustained	-16.25	0.5695
methods	20	237	Founder Sustained	Later Joiner Sustained	-13.49	0.2351

Post-hoc Pairwise Analysis: Holm-Bonferroni Correction

To evaluate specific distribution differences between isolated developer interaction groups without violating normality or homoscedastic constraints, we implement a two-sample rank-order framework adjusted by the sequentially rejective Holm method.

The raw P-values for a set of m concurrent hypotheses are sorted in ascending order:

$$P_{(1)} \leq P_2 \leq \cdots \leq P_{(m)} \tag{3}$$

Each index i $(1 \leq i \leq m)$ is evaluated sequentially against a dynamically scaled significance threshold. The condition for rejecting the local null hypothesis $H_{(i)}$ is defined as:

$$P_{(i)} \leq \frac{\alpha}{m - i + 1} \tag{4}$$

Where:

- α is the nominal family-wise error criteria cutoff (configured to 0.05).
- m is the total number of simultaneous pairwise comparisons evaluated.
- i is the ordered ranking position of the active sub-test segment.

The sequence terminates at the first instance where $P_{(i)} > \frac{\alpha}{m-i+1}$, retaining all remaining null structures.

9 Strategy Logistic Regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **30% Threshold:** The threshold number of components to touch to achieve a 30% empirical probability of transitioning into a sustained contributor.
- **Not Sustained Met Threshold:** The not Sustained number of developers to have met the threshold, the percentage of developers in brackets.
- **Sustained Met Threshold:** The Sustained number of developers to have met the threshold, the percentage of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 18: Strategy Logistic Regression for all components against number of commits to predict Sustained developers

Component	Number of First Commits	Model Parameter	Number of Developers	Not Sustained μ	Sustained μ	Coefficient	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Sustained Met Threshold	Sustained Met Threshold
packages	1	Intercept	16217	1.21	2.38	-4.4747	0.0114	0.0772	0.0000	-	-	-
packages	1	Packages Count	16217	1.21	2.38	0.1789	1.1959	0.0239	0.0000	**20.2798**	6 (0.04%)	1 (0.42%)
packages	2	Intercept	3088	2.19	2.38	-2.5423	0.0787	0.0855	0.0000	-	-	-
packages	2	Packages Count	3088	2.19	2.38	0.0241	1.0244	0.0223	0.2783	**70.1988**	0 (0.00%)	0 (0.00%)
packages	5	Intercept	1363	2.39	2.38	-1.5543	0.2113	0.0970	0.0000	-	-	-
packages	5	Packages Count	1363	2.39	2.38	-0.0017	0.9983	0.0276	0.9502	**-410.1531**	1126 (100.00%)	237 (100.00%)
packages	10	Intercept	858	2.55	2.38	-0.9027	0.4055	0.1058	0.0000	-	-	-
packages	10	Packages Count	858	2.55	2.38	-0.0246	0.9757	0.0303	0.4167	**-2.2484**	621 (100.00%)	237 (100.00%)
classes	1	Classes Count	16217	1.24	5.19	0.0397	1.0405	0.0071	0.0000	**86.4559**	5 (0.03%)	2 (0.84%)
classes	2	Intercept	3088	2.36	5.19	-2.5310	0.0796	0.0703	0.0000	-	-	-
classes	2	Classes Count	3088	2.36	5.19	0.0142	1.0143	0.0058	0.0151	**118.4255**	1 (0.04%)	2 (0.84%)
classes	5	Intercept	1363	2.77	5.19	-1.5917	0.2036	0.0741	0.0000	-	-	-
classes	5	Classes Count	1363	2.77	5.19	0.0096	1.0096	0.0056	0.0872	**77.7562**	4 (0.36%)	2 (0.84%)
classes	10	Intercept	858	3.30	5.19	-0.9855	0.3732	0.0785	0.0000	-	-	-
classes	10	Classes Count	858	3.30	5.19	0.0056	1.0056	0.0046	0.2207	**24.7871**	13 (2.09%)	6 (2.53%)
methods	1	Methods Count	16217	1.80	22.14	0.0068	1.0068	0.0015	0.0000	**501.2763**	2 (0.01%)	2 (0.84%)
methods	2	Intercept	3088	5.48	22.14	-2.5063	0.0816	0.0687	0.0000	-	-	-
methods	2	Methods Count	3088	5.48	22.14	0.0022	1.0022	0.0014	0.1213	**745.5317**	1 (0.04%)	1 (0.42%)
methods	5	Intercept	1363	6.89	22.14	-1.5747	0.2071	0.0726	0.0000	-	-	-
methods	5	Methods Count	1363	6.89	22.14	0.0016	1.0016	0.0014	0.2568	**454.4624**	2 (0.18%)	2 (0.84%)
methods	10	Intercept	858	8.91	22.14	-0.9754	0.3771	0.0771	0.0000	-	-	-
methods	10	Methods Count	858	8.91	22.14	0.0009	1.0009	0.0009	0.2850	**135.7767**	6 (0.97%)	3 (1.27%)

Table 19: Strategy Logistic Regression for all components against number of commits to predict Core developers

Component	Number of First Commits	Model Parameter	Number of Developers	Not Core μ	Core μ	Coefficient	Odds Ratio	Std. Err	p-Value	30.0% Threshold	Not Core Met Threshold	Core Met Threshold
packages	1	Intercept	16217	1.22	1.84	-4.5484	0.0106	0.0799	0.0000	-	-	-
packages	1	Packages Count	16217	1.22	1.84	0.1115	1.1179	0.0219	0.0000	**33.2067**	3 (0.02%)	0 (0.00%)
packages	2	Intercept	3088	2.19	2.49	-3.3587	0.0348	0.1182	0.0000	-	-	-
packages	2	Packages Count	3088	2.19	2.49	0.0339	1.0345	0.0278	0.2228	**74.0388**	0 (0.00%)	0 (0.00%)
packages	5	Intercept	1363	2.37	2.51	-2.5068	0.0815	0.1318	0.0000	-	-	-
packages	5	Packages Count	1363	2.37	2.51	0.0180	1.0182	0.0340	0.5963	**92.0840**	0 (0.00%)	0 (0.00%)
packages	10	Intercept	858	2.51	2.48	-1.9731	0.1390	0.1409	0.0000	-	-	-
packages	10	Packages Count	858	2.51	2.48	-0.0032	0.9968	0.0380	0.9334	**-354.5509**	754 (100.00%)	104 (100.00%)
packages	20	Intercept	237	2.29	2.64	-1.2383	0.2899	0.2010	0.0000	-	-	-
packages	20	Packages Count	237	2.29	2.64	0.0455	1.0466	0.0528	0.3884	**8.5880**	4 (2.23%)	2 (3.45%)
classes	1	Intercept	16217	1.27	3.67	-4.4218	0.0120	0.0728	0.0000	-	-	-
classes	1	Classes Count	16217	1.27	3.67	0.0169	1.0170	0.0073	0.0198	**211.5752**	1 (0.01%)	0 (0.00%)
classes	2	Intercept	3088	2.45	5.72	-3.3121	0.0364	0.0980	0.0000	-	-	-
classes	2	Classes Count	3088	2.45	5.72	0.0097	1.0097	0.0045	0.0316	**254.4410**	1 (0.03%)	0 (0.00%)
classes	5	Intercept	1363	2.96	5.90	-2.4901	0.0829	0.1027	0.0000	-	-	-
classes	5	Classes Count	1363	2.96	5.90	0.0070	1.0070	0.0044	0.1074	**234.5535**	1 (0.08%)	0 (0.00%)
classes	10	Intercept	858	3.55	5.78	-2.0011	0.1352	0.1067	0.0000	-	-	-
classes	10	Classes Count	858	3.55	5.78	0.0046	1.0046	0.0042	0.2752	**251.6201**	1 (0.13%)	0 (0.00%)
classes	20	Intercept	237	4.57	7.12	-1.1404	0.3197	0.1535	0.0000	-	-	-
classes	20	Classes Count	237	4.57	7.12	0.0024	1.0024	0.0044	0.5874	**122.9235**	1 (0.56%)	1 (1.72%)
methods	1	Intercept	16217	2.01	9.51	-4.3980	0.0123	0.0716	0.0000	-	-	-
methods	1	Methods Count	16217	2.01	9.51	0.0013	1.0013	0.0006	0.0372	**2818.7606**	1 (0.01%)	0 (0.00%)
methods	2	Intercept	3088	6.41	16.04	-3.2872	0.0374	0.0966	0.0000	-	-	-
methods	2	Methods Count	3088	6.41	16.04	0.0008	1.0008	0.0006	0.2189	**3142.4050**	1 (0.03%)	0 (0.00%)
methods	5	Intercept	1363	8.97	16.27	-2.4684	0.0847	0.1012	0.0000	-	-	-
methods	5	Methods Count	1363	8.97	16.27	0.0005	1.0005	0.0007	0.4789	**3474.0794**	0 (0.00%)	0 (0.00%)
methods	10	Intercept	858	12.07	16.16	-1.9842	0.1375	0.1052	0.0000	-	-	-
methods	10	Methods Count	858	12.07	16.16	0.0002	1.0002	0.0007	0.7454	**4981.3682**	0 (0.00%)	0 (0.00%)
methods	20	Intercept	237	23.43	18.17	-1.1244	0.3248	0.1518	0.0000	-	-	-
methods	20	Methods Count	237	23.43	18.17	-0.0001	0.9999	0.0008	0.8737	**-2243.3341**	179 (100.00%)	58 (100.00%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n \tag{5}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{6}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy[cite: 1639].

10 Granular Exploration Profiles: Evaluating the Method-to-Package Ratio

To evaluate the operational strategies employed by developers when entering a codebase, we analyse the structural ratio of unique methods modified relative to unique packages touched:

$$\text{Ratio}_{m/p} = \frac{\text{Methods Touched}}{\text{Packages Touched}} \tag{7}$$

This ratio serves as a quantitative proxy for codebase exploration tactics during a contributor's initial commits. We categorize exploration strategies using empirical threshold boundaries:

- **Breadth-First Strategy ($\textbf{Ratio}_{m/p} < 25$):** Indicates a structural approach where edits are distributed across package boundaries without heavy localized method churn.
- **Moderate Exploration ($25 \leq \textbf{Ratio}_{m/p} \leq 100$):** Represents a balanced mix of architectural traversal and module-level implementation.
- **Depth-First Strategy ($\textbf{Ratio}_{m/p} > 100$):** Signifies a highly localized strategy where engineering effort is concentrated deep within a narrow set of architectural packages.

To test whether sustained developers systematically adopt a predominant strategy or bifurcate into distinct archetypes, we execute Hartigan's Dip Test for multimodality alongside two-sample non-parametric tests (Mann-Whitney U and Brunner–Münzel).

Table 20: Method-to-Package Exploration Strategy Distribution and Multimodality Test

Commits	**Sustained Devs**	**Median** $Ratio_{m/p}$	**Breadth-First** (< 25)	**Depth-First** (> 100)	**Dip Stat** (p-**val**)	**MWU** p-**val**	**Binomial** p-**val**
1	237	1.00	232 (97.9%)	2 (0.8%)	0.8850 (0.0450)	0.0214	0.0000
2	237	2.17	217 (91.6%)	6 (2.5%)	0.5951 (0.0450)	0.0805	0.0000
5	237	8.00	186 (78.5%)	12 (5.1%)	0.4874 (0.0450)	0.0566	0.0000
10	237	15.00	164 (69.2%)	24 (10.1%)	0.4802 (0.0450)	0.0078	0.0000
20	237	20.60	134 (56.5%)	28 (11.8%)	0.3311 (0.0450)	nan	0.0000

11 Repository

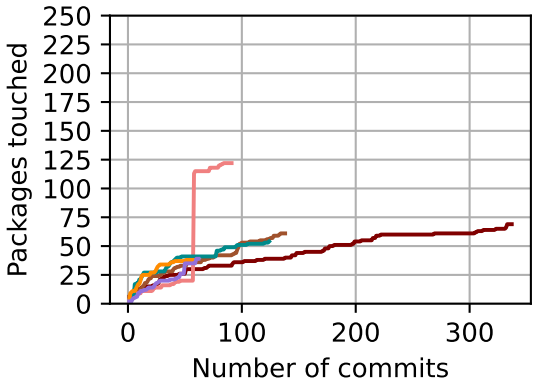
11.1 Repository: 14

Table 21: Table of first 6 sustained founder developers in the Scatter Developer graphs.

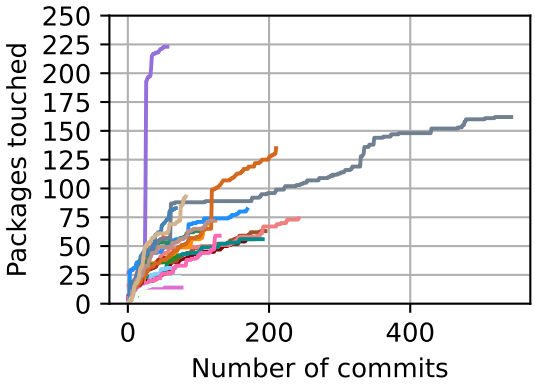
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 22: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

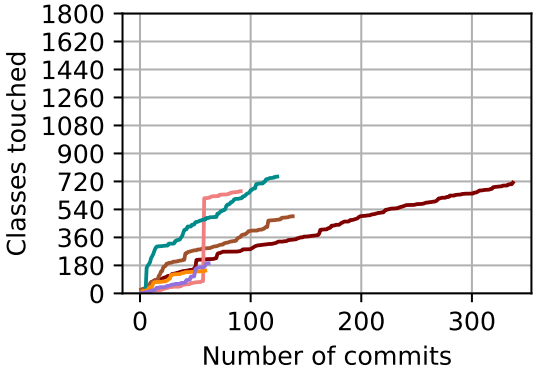
Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



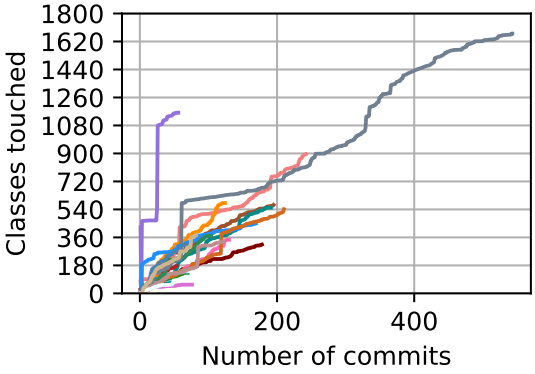
(a) Packages for sustained founder



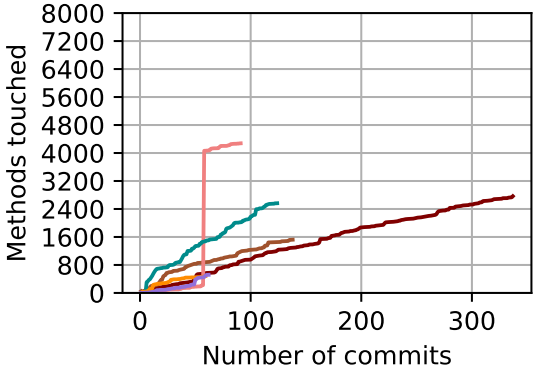
(b) Packages for sustained later joiner



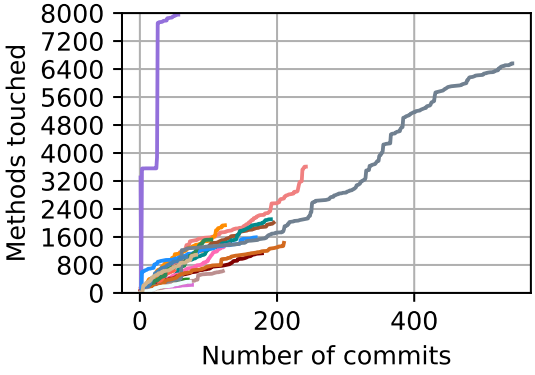
(c) Classes for sustained founder



(d) Classes for sustained later joiner

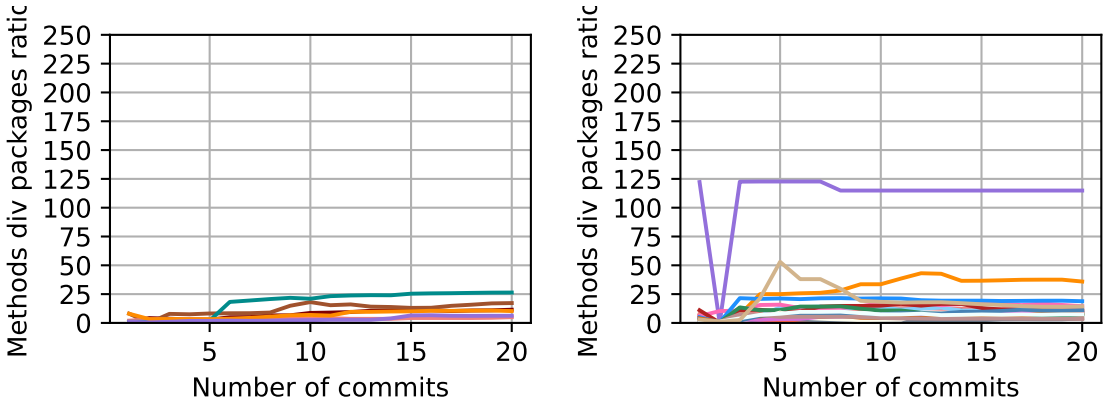


(e) Methods for sustained founder



(f) Methods for sustained later joiner

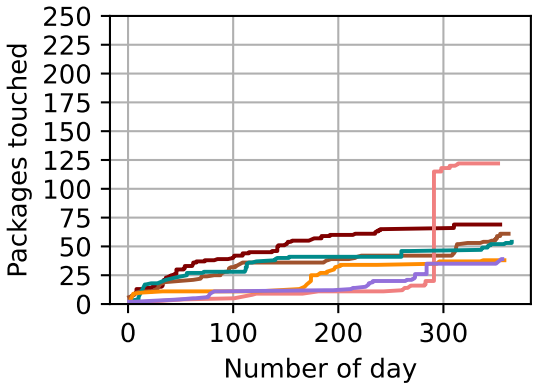
Figure 9: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 21. Sustained later joiner developers with colours are shown in Table 22.



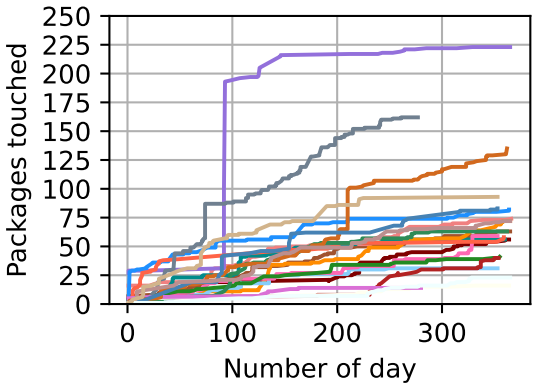
(a) Methods for sustained founder

(b) Methods for sustained later joiner

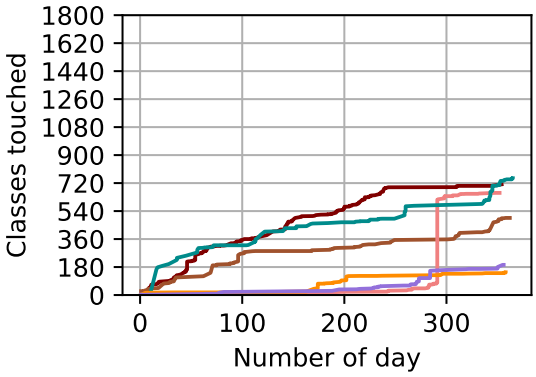
Figure 10: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 21. Sustained later joiner developers with colours are shown in Table 22.



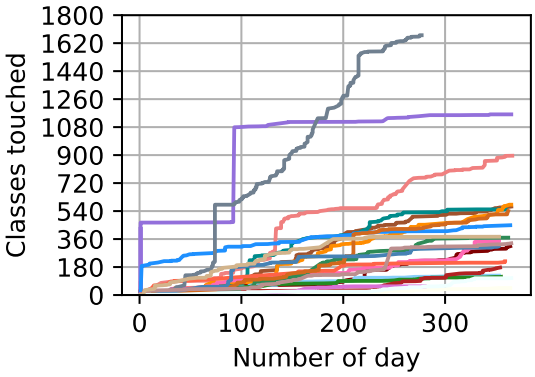
(a) Packages for sustained founder



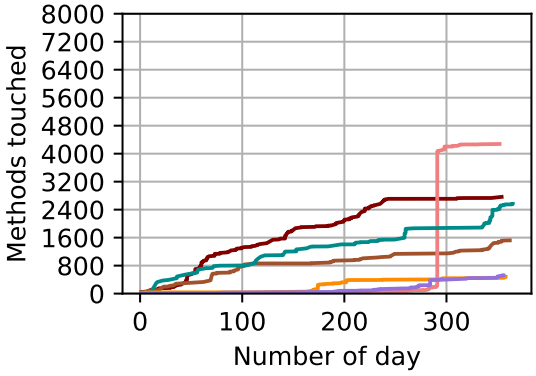
(b) Packages for sustained later joiner



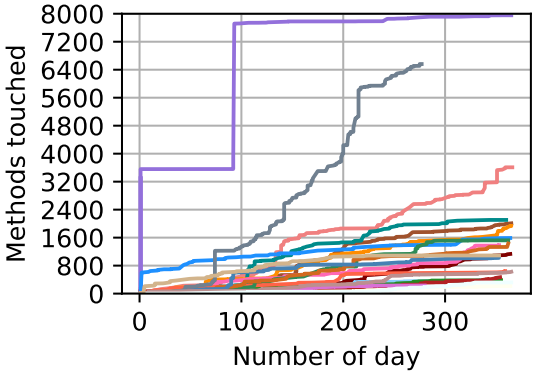
(c) Classes for sustained founder



(d) Classes for sustained later joiner

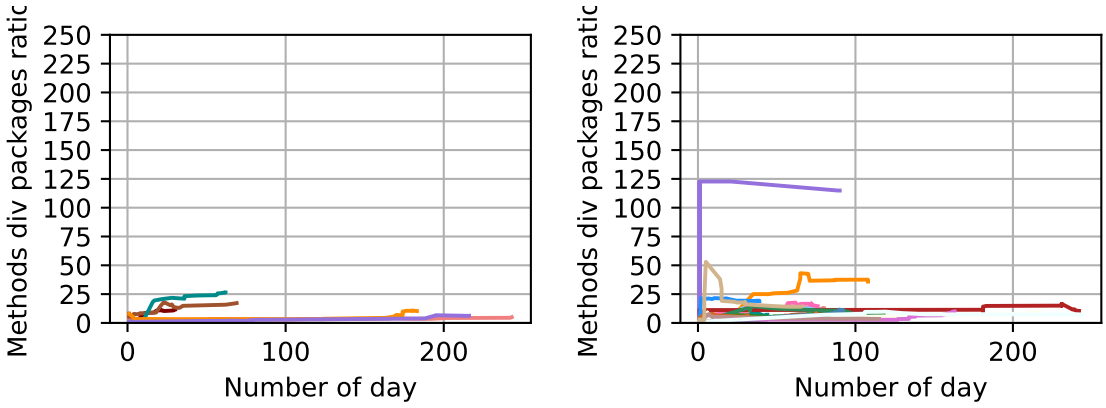


(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 11: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 21. Sustained later joiner developers with colours are shown in Table 22.



(a) Methods for sustained founder

(b) Methods for sustained later joiner

Figure 12: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 21. Sustained later joiner developers with colours are shown in Table 22.

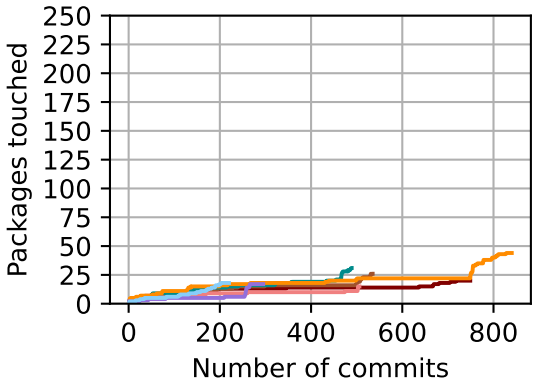
11.2 Repository: 21

Table 23: Table of first 7 sustained founder developers in the Scatter Developer graphs.

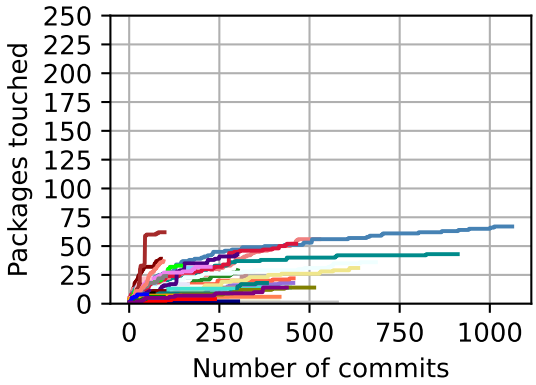
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 24: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

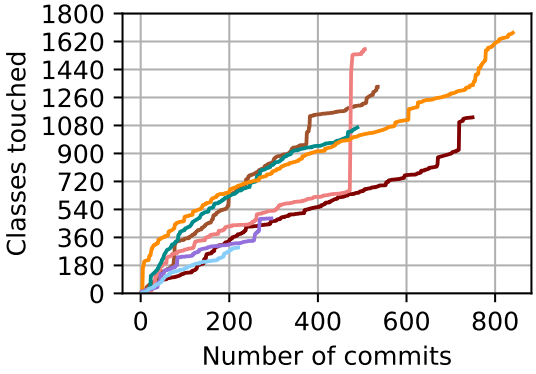
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



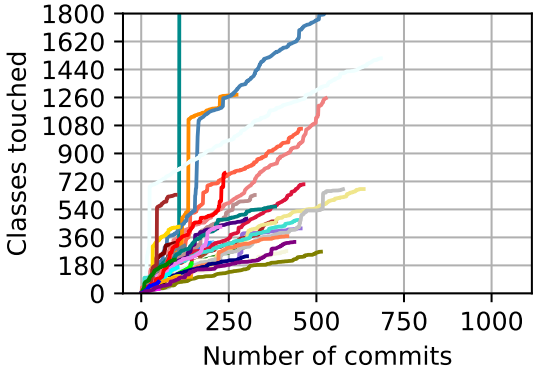
(a) Packages for sustained founder



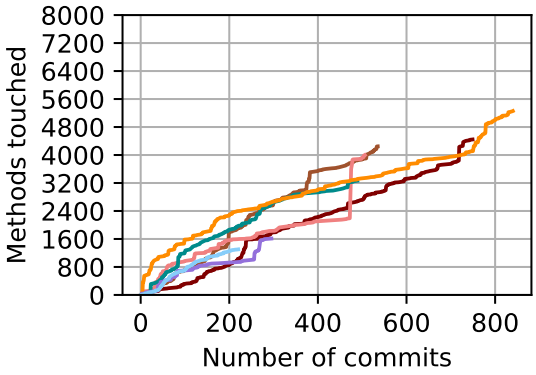
(b) Packages for sustained later joiner



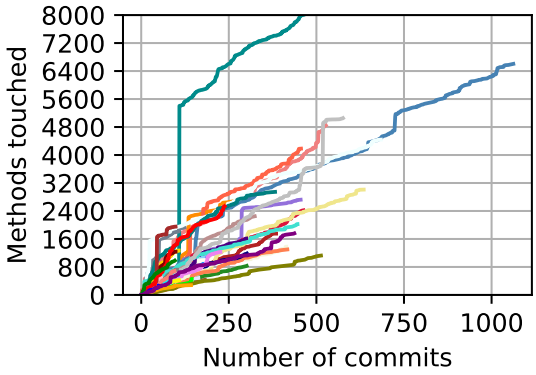
(c) Classes for sustained founder



(d) Classes for sustained later joiner

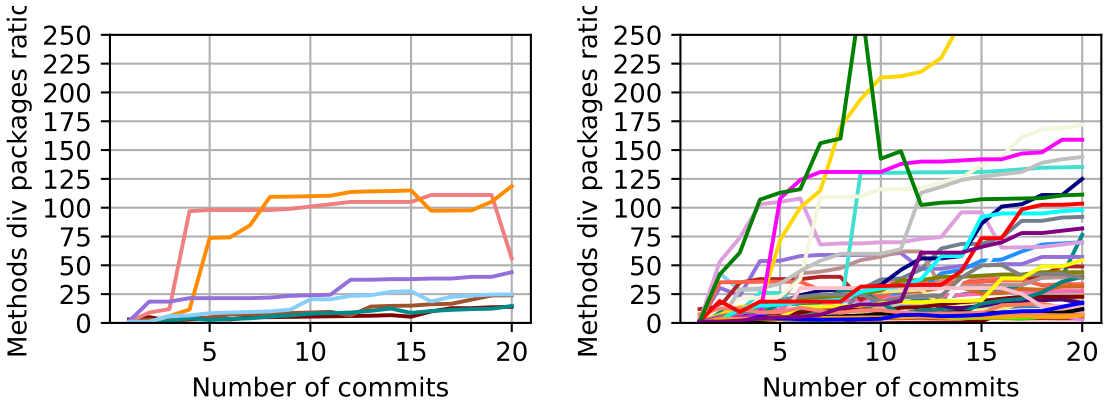


(e) Methods for sustained founder



(f) Methods for sustained later joiner

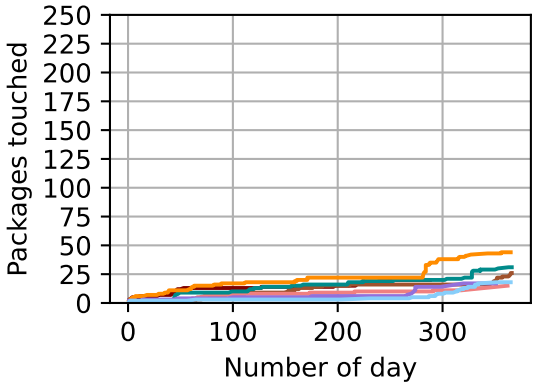
Figure 13: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 23. Sustained later joiner developers with colours are shown in Table 24.



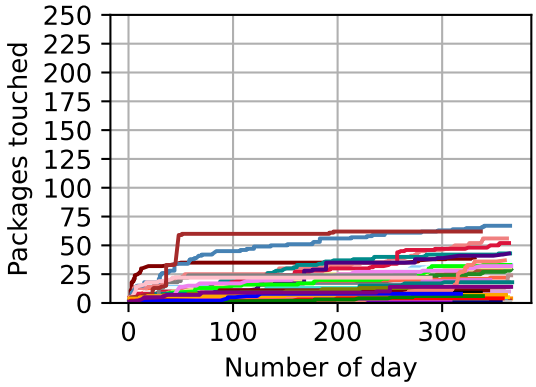
(a) Methods for sustained founder

(b) Methods for sustained later joiner

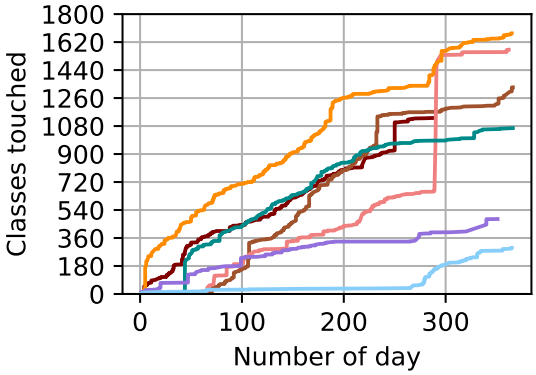
Figure 14: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 23. Sustained later joiner developers with colours are shown in Table 24.



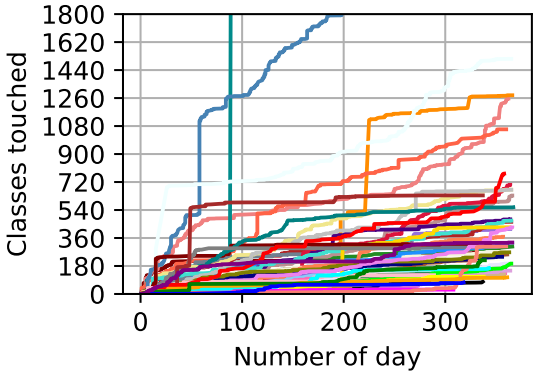
(a) Packages for sustained founder



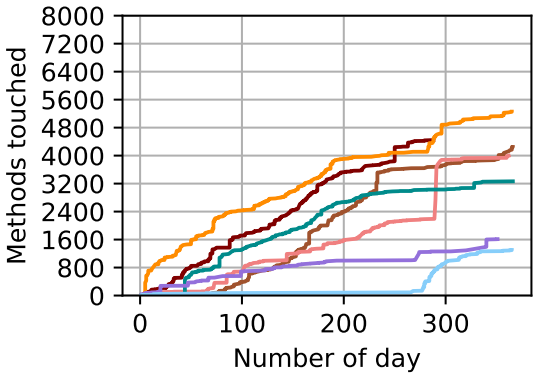
(b) Packages for sustained later joiner



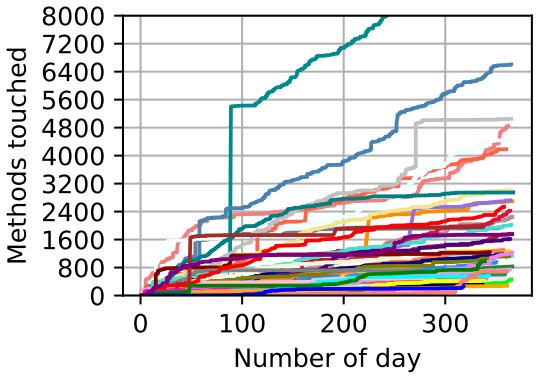
(c) Classes for sustained founder



(d) Classes for sustained later joiner

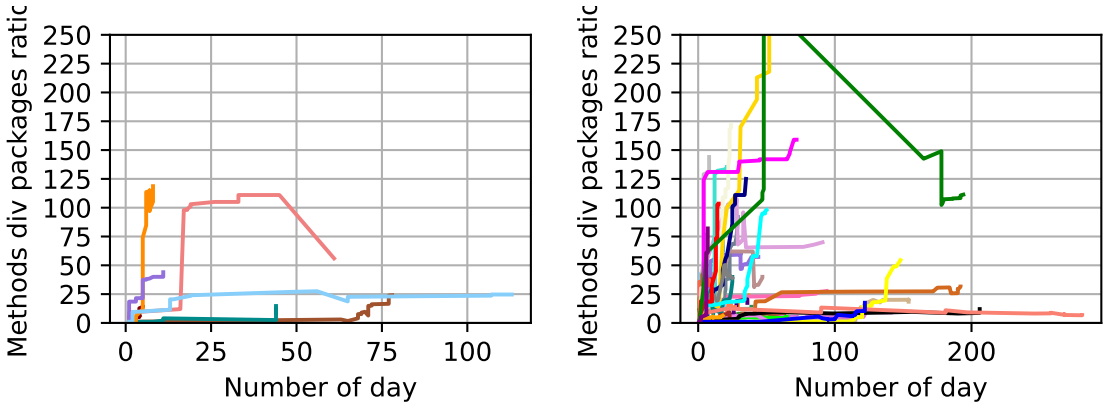


(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 15: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 23. Sustained later joiner developers with colours are shown in Table 24.



(a) Methods for sustained founder

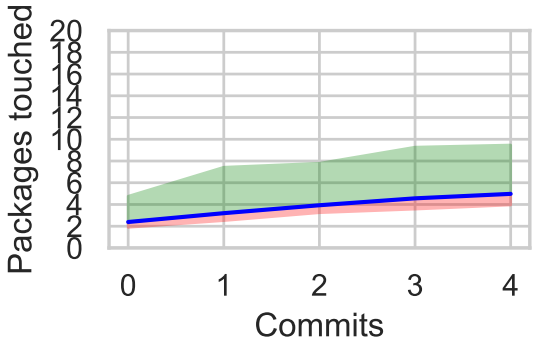
(b) Methods for sustained later joiner

Figure 16: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components methods div packages ratio (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 23. Sustained later joiner developers with colours are shown in Table 24.

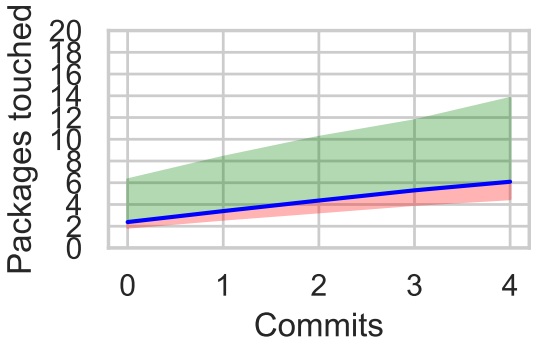
Appendices

A Time Series Components Touched - For all components touched for all commits and for each period

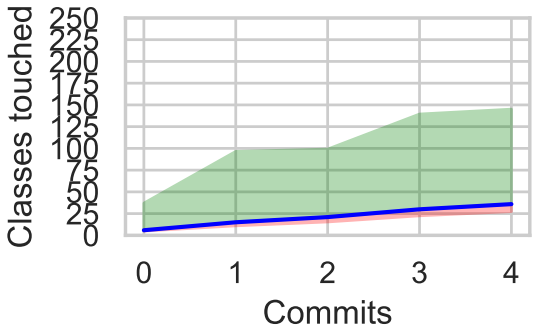
A.1 Time Series Components Touched - For packages touched for first five commits



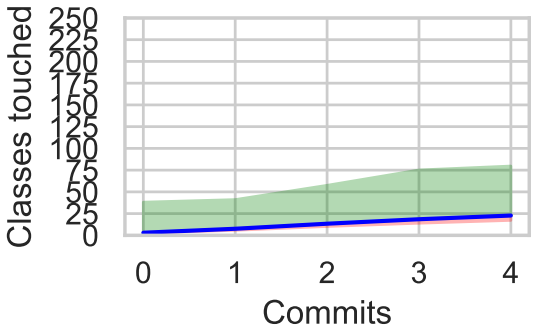
(a) Packages for All founders (n=85, μ=4.98, median=3.0 & std=3.56)



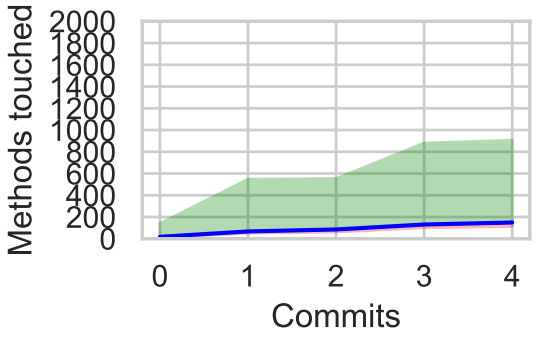
(b) Packages for All joiners (n=1277, μ=6.09, median=3.0 & std=4.66)



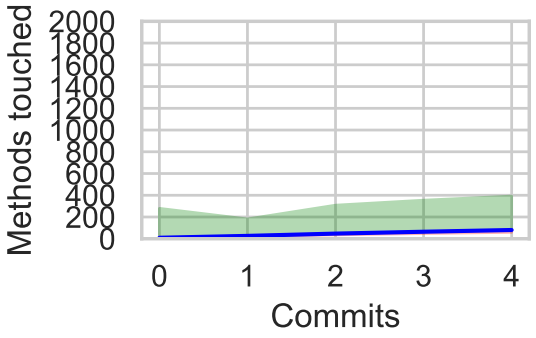
(c) Classes for All founders (n=85, μ=35.89, median=7.0 & std=51.26)



(d) Classes for All joiners (n=1277, μ=22.77, median=6.0 & std=28.25)

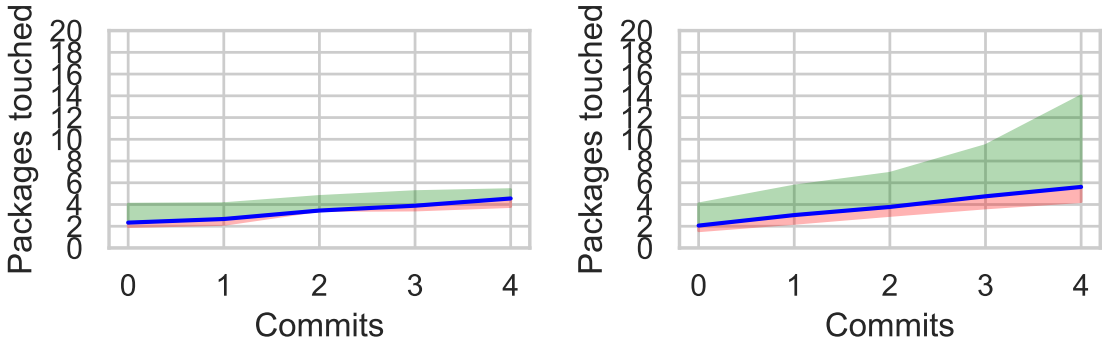


(e) Methods for All founders (n=85, μ=149.4, median=17.0 & std=292.52)



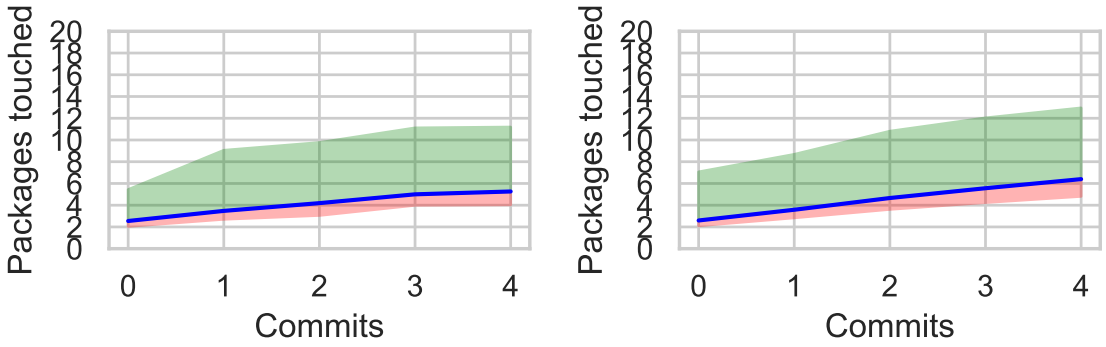
(f) Methods for All joiners (n=1277, μ=80.35, median=12.0 & std=138.17)

Figure 17: A time series for the five (5) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



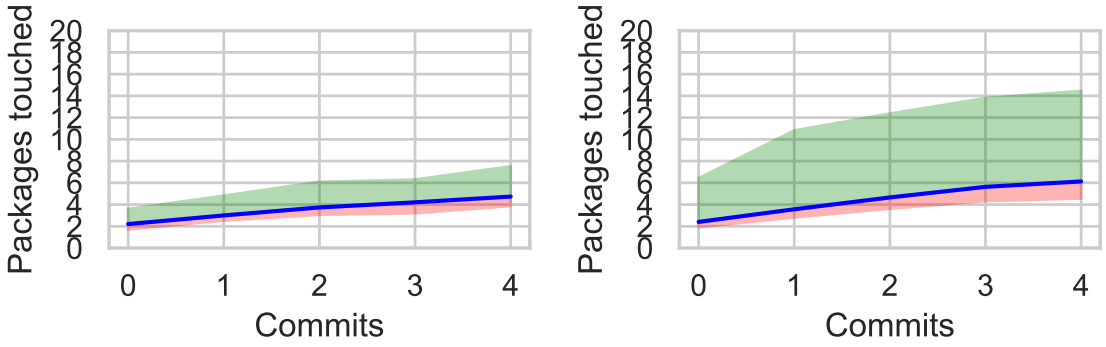
(a) Transient founder (n=9, μ=4.56, median=3.0 & std=1.99)

(b) Transient later joiner (n=429, μ=5.62, median=3.0 & std=3.96)



(c) Moderate founder (n=42, μ=5.26, median=3.0 & std=4.29)

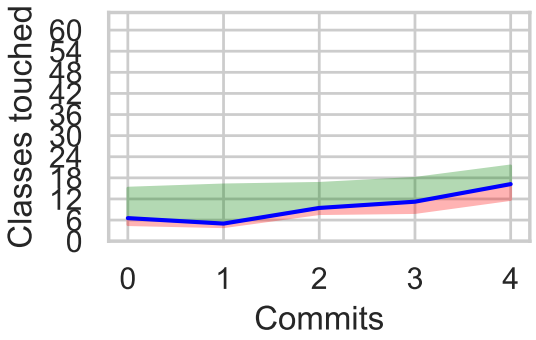
(d) Moderate later joiner (n=646, μ=6.4, median=3.0 & std=4.82)



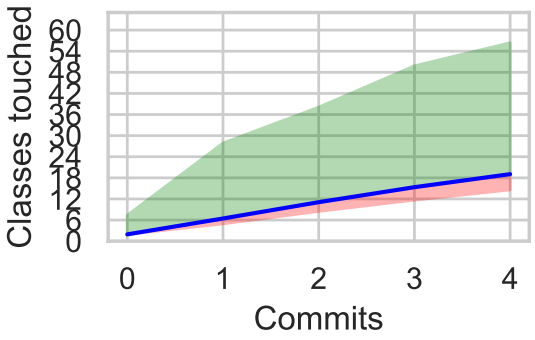
(e) Sustained founder (n=34, μ=4.74, median=2.0 & std=2.79)

(f) Sustained later joiner (n=203, μ=6.13, median=3.0 & std=5.36)

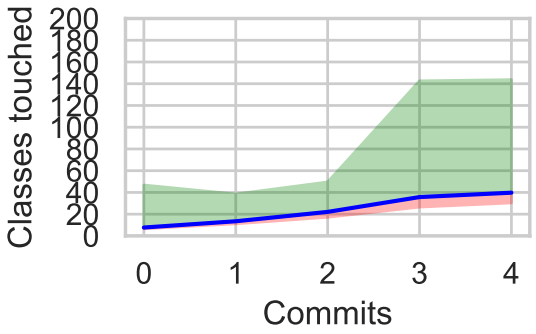
Figure 18: A time series for the five (5) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



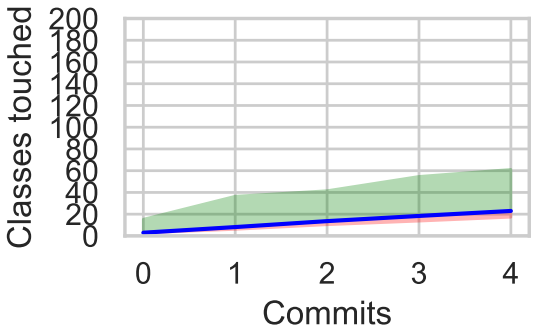
(a) Transient founder (n=9, μ=16.22, median=6.0 & std=9.9)



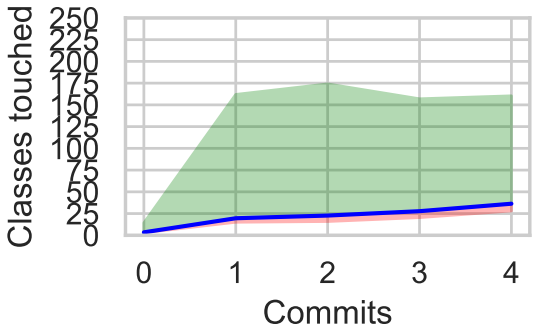
(b) Transient later joiner (n=429, μ=19.05, median=5.0 & std=19.33)



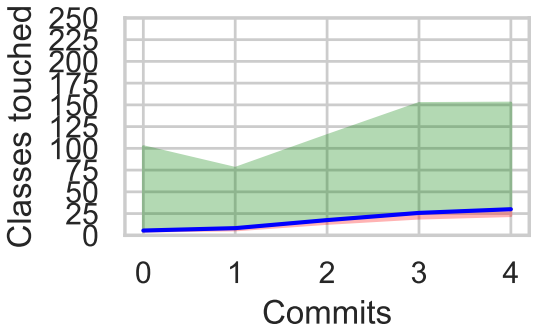
(c) Moderate founder (n=42, μ=39.76, median=9.0 & std=48.23)



(d) Moderate later joiner (n=646, μ=22.96, median=6.0 & std=21.93)

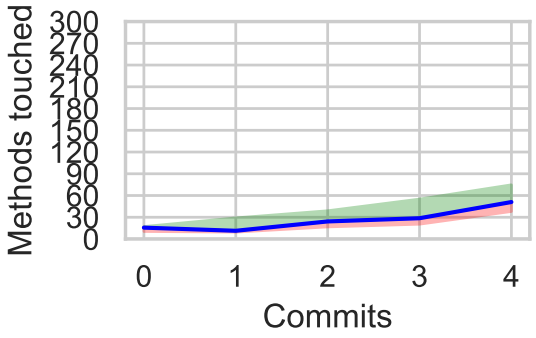


(e) Sustained founder (n=34, μ=36.32, median=7.0 & std=60.21)

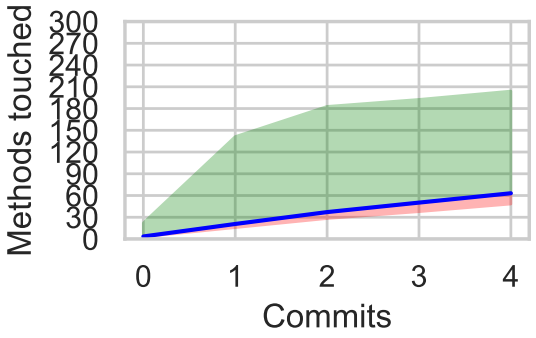


(f) Sustained later joiner (n=203, μ=30.01, median=6.0 & std=51.67)

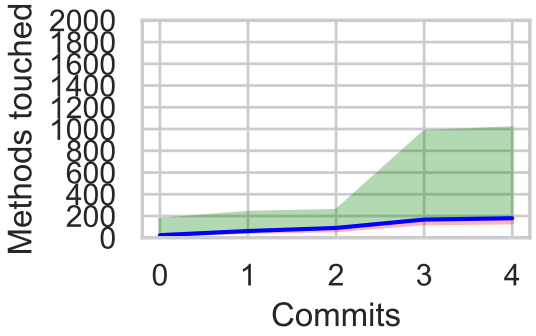
Figure 19: A time series for the five (5) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



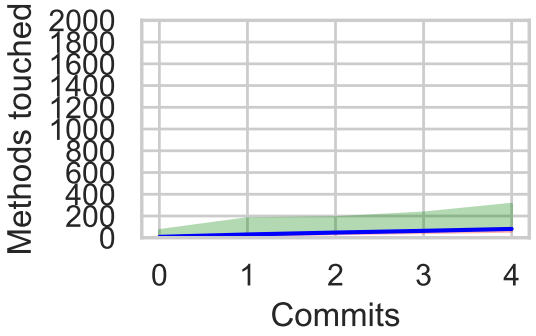
(a) Transient founder (n=9, μ=51.0, median=16.0 & std=30.34)



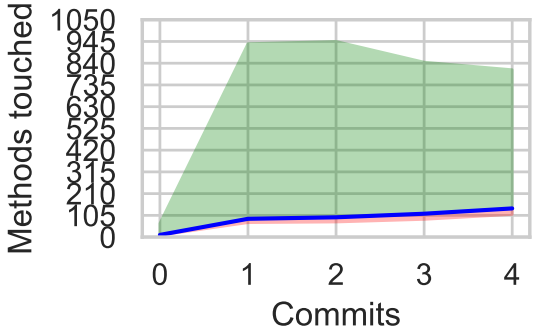
(b) Transient later joiner (n=429, μ=63.11, median=10.0 & std=79.44)



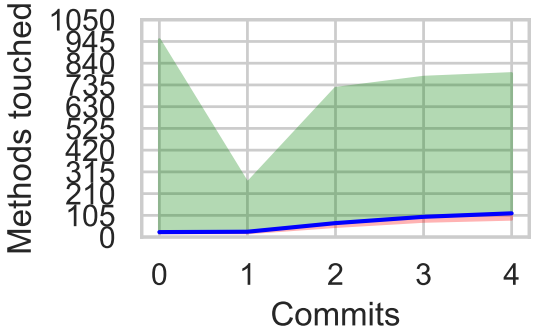
(c) Moderate founder (n=42, μ=179.29, median=21.5 & std=305.37)



(d) Moderate later joiner (n=646, μ=80.93, median=14.5 & std=100.84)



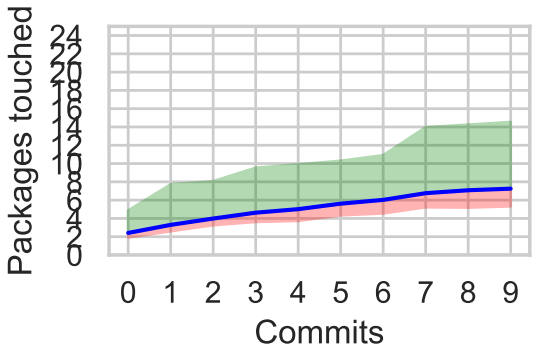
(e) Sustained founder (n=34, μ=138.53, median=15.0 & std=311.7)



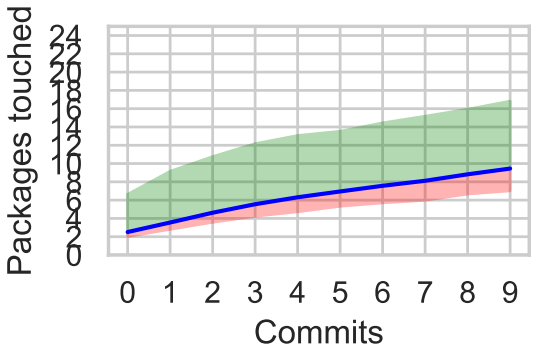
(f) Sustained later joiner (n=203, μ=114.77, median=13.0 & std=271.55)

Figure 20: A time series for the five (5) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for six (6) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

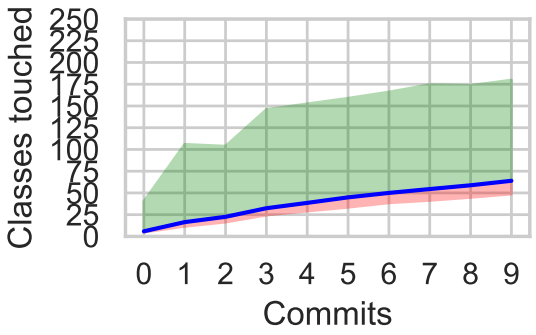
A.2 Time Series Components Touched - For packages touched for first ten commits



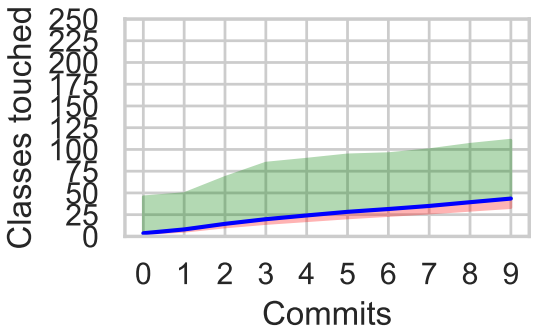
(a) Packages for All founders (n=75, μ=7.25, median=4.0 & std=5.2)



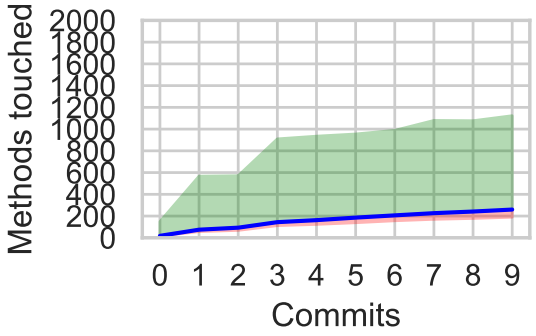
(b) Packages for All joiners (n=783, μ=9.45, median=5.0 & std=6.16)



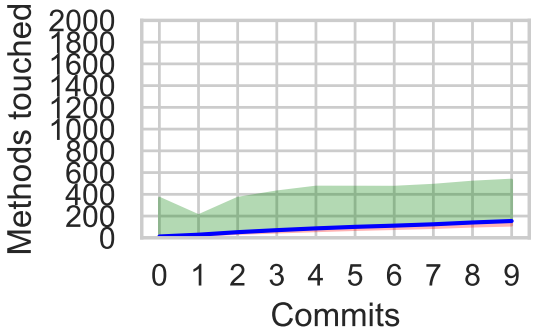
(c) Classes for All founders (n=75, μ=63.96, median=16.5 & std=71.03)



(d) Classes for All joiners (n=783, μ=43.51, median=14.0 & std=40.75)

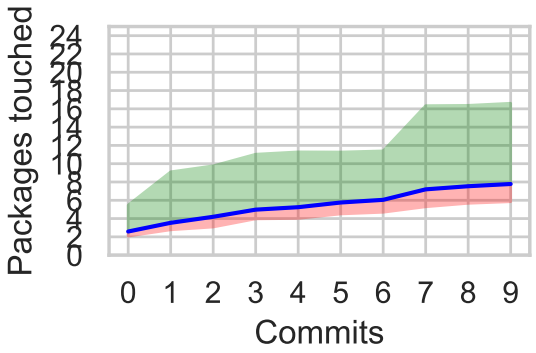


(e) Methods for All founders (n=75, μ=259.76, median=44.0 & std=414.98)

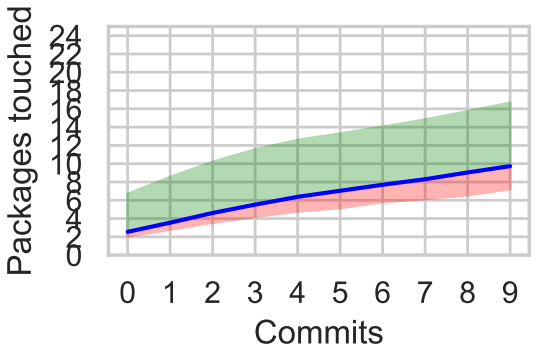


(f) Methods for All joiners (n=783, μ=154.41, median=37.0 & std=201.73)

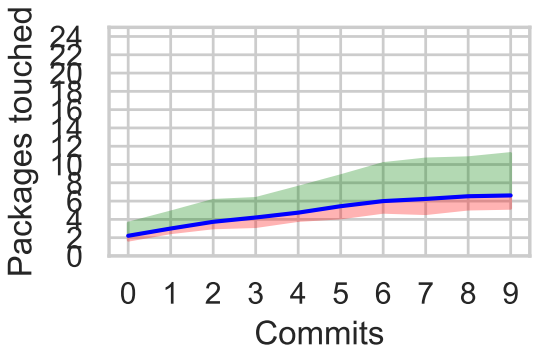
Figure 21: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



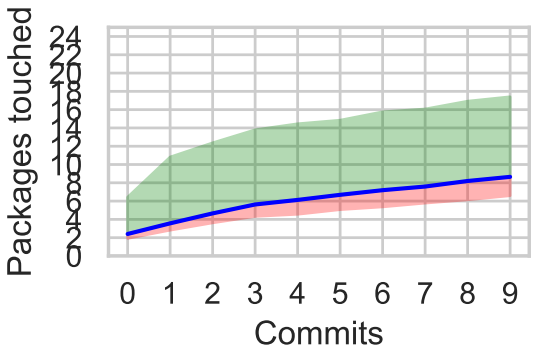
(a) Moderate founder (n=41, μ=7.78, median=4.0 & std=5.87)



(b) Moderate later joiner (n=580, μ=9.73, median=5.0 & std=6.0)

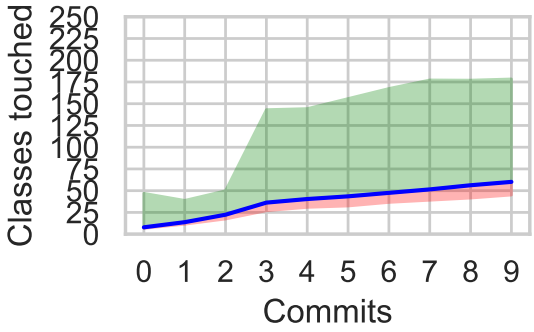


(c) Sustained founder (n=34, μ=6.62, median=3.0 & std=4.25)

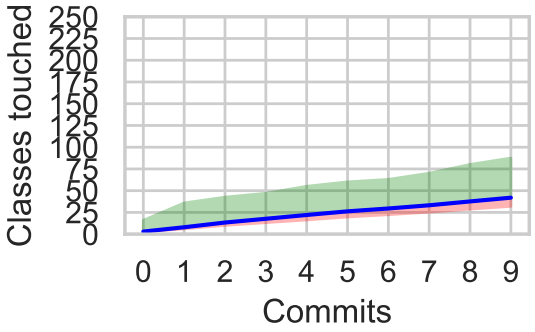


(d) Sustained later joiner (n=203, μ=8.65, median=4.0 & std=6.6)

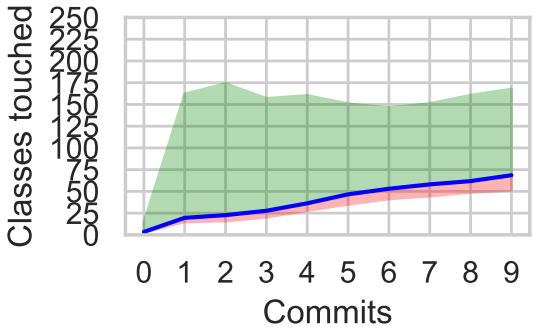
Figure 22: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



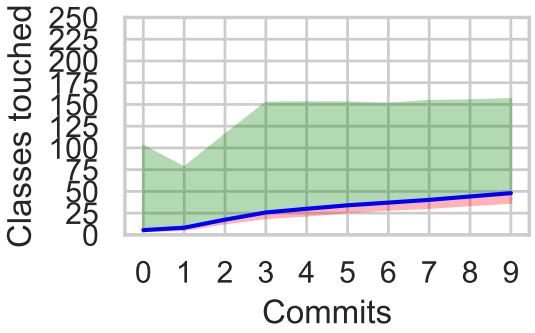
(a) Moderate founder (n=41, μ=60.12, median=18.0 & std=69.27)



(b) Moderate later joiner (n=580, μ=41.95, median=14.0 & std=29.54)

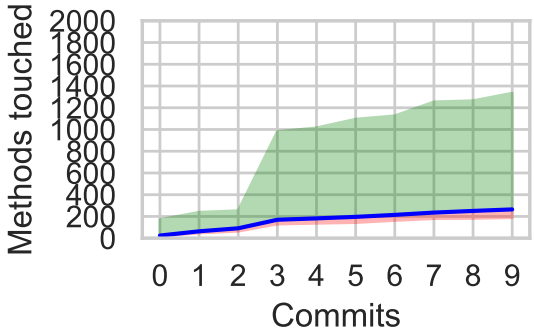


(c) Sustained founder (n=34, μ=68.59, median=15.0 & std=73.09)

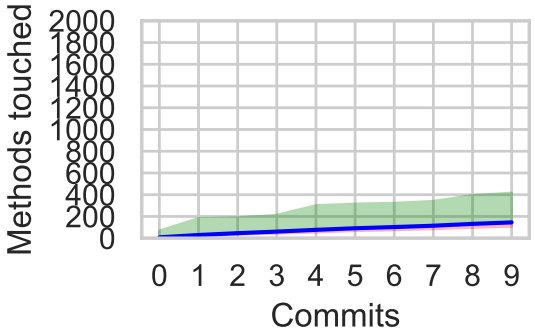


(d) Sustained later joiner (n=203, μ=47.98, median=13.0 & std=62.35)

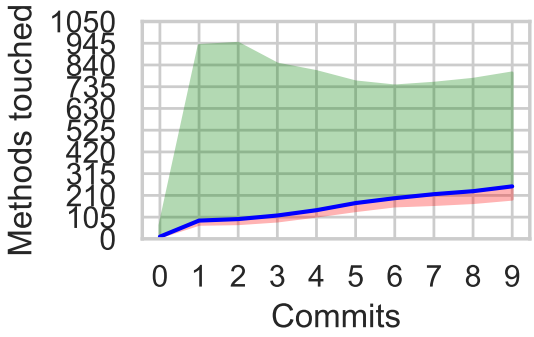
Figure 23: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



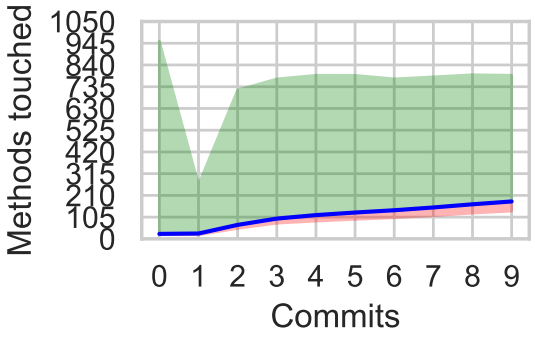
(a) Moderate founder (n=41, μ=264.83, median=46.5 & std=461.01)



(b) Moderate later joiner (n=580, μ=145.21, median=38.0 & std=142.87)



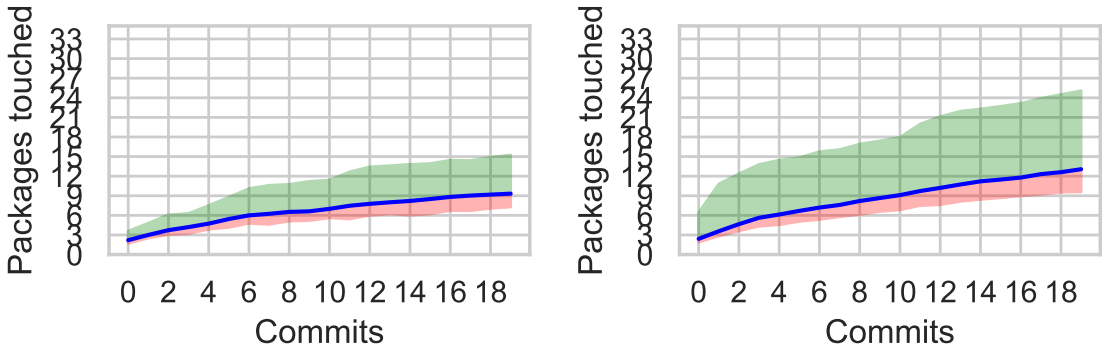
(c) Sustained founder (n=34, μ=253.65, median=42.5 & std=351.3)



(d) Sustained later joiner (n=203, μ=180.7, median=34.0 & std=313.08)

Figure 24: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

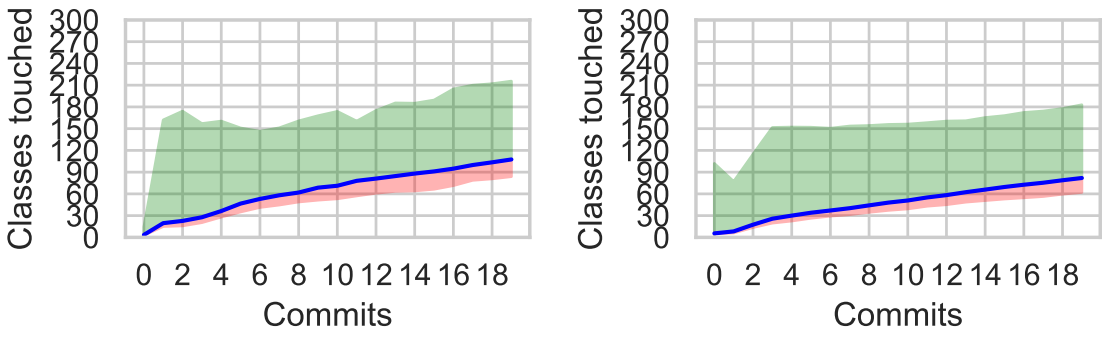
A.3 Time Series Components Touched - For packages touched for first twenty commits



(a) Sustained founder (n=34, μ=9.32, median=4.0 & std=6.14)

(b) Sustained later joiner (n=203, μ=13.06, median=6.0 & std=9.14)

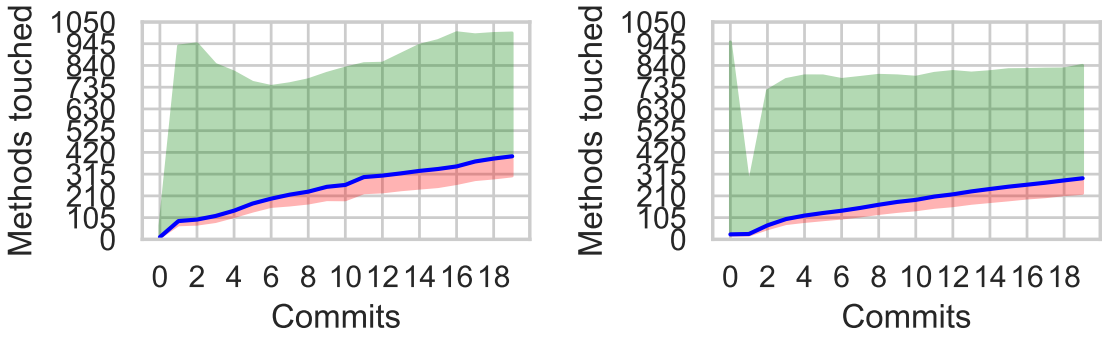
Figure 25: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=107.5, median=30.0 & std=91.26)

(b) Sustained later joiner (n=203, μ=81.91, median=27.0 & std=72.01)

Figure 26: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

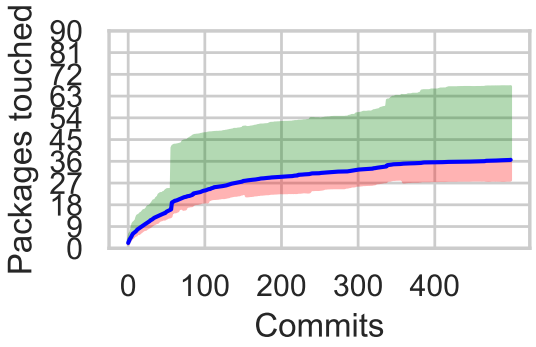


(a) Sustained founder (n=34, μ=401.24, median=85.5 & std=426.09)

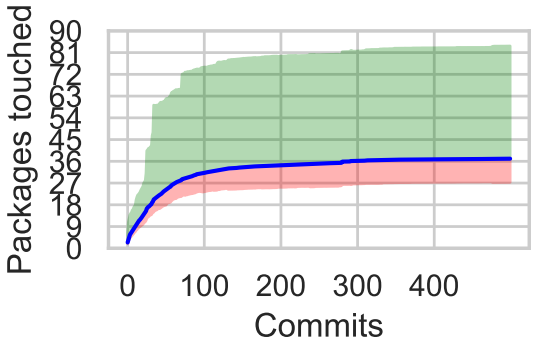
(b) Sustained later joiner (n=203, μ=295.1, median=79.0 & std=349.63)

Figure 27: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

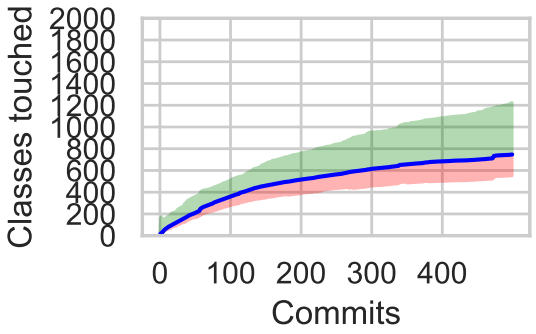
A.4 Time Series Components Touched - For packages touched for commit



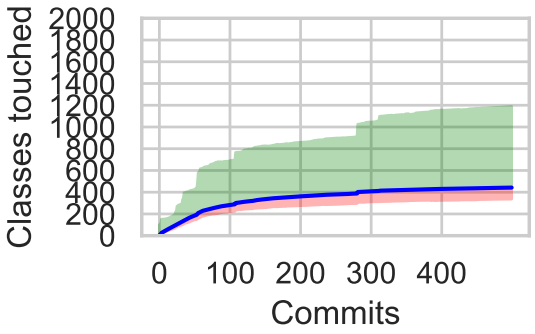
(a) Packages for All founders (n=34, μ=36.62, median=18.0 & std=28.13)



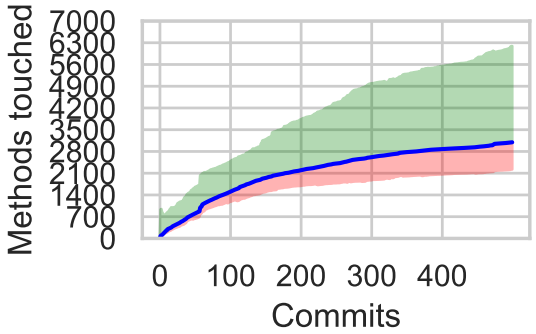
(b) Packages for All joiners (n=203, μ=37.08, median=22.0 & std=36.82)



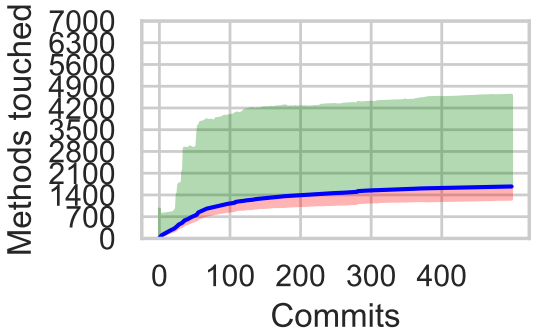
(c) Classes for All founders (n=34, μ=746.59, median=426.0 & std=403.07)



(d) Classes for All joiners (n=203, μ=442.33, median=239.0 & std=426.23)

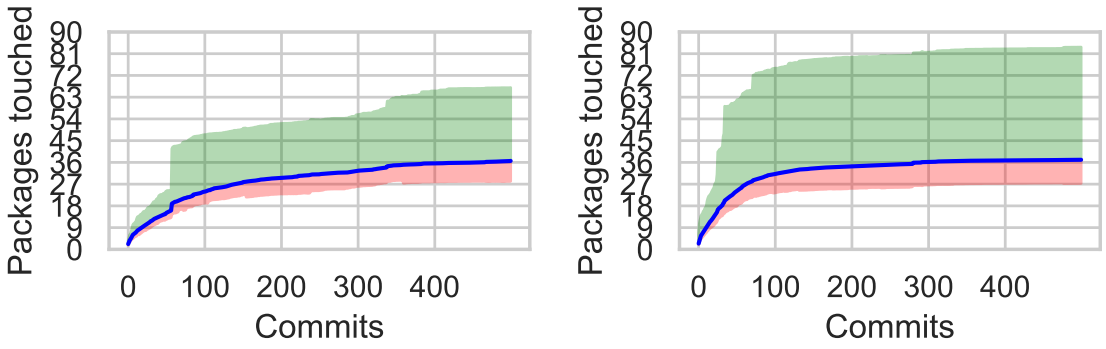


(e) Methods for All founders (n=34, μ=3095.18, median=1645.5 & std=2043.47)



(f) Methods for All joiners (n=203, μ=1672.84, median=838.0 & std=1909.9)

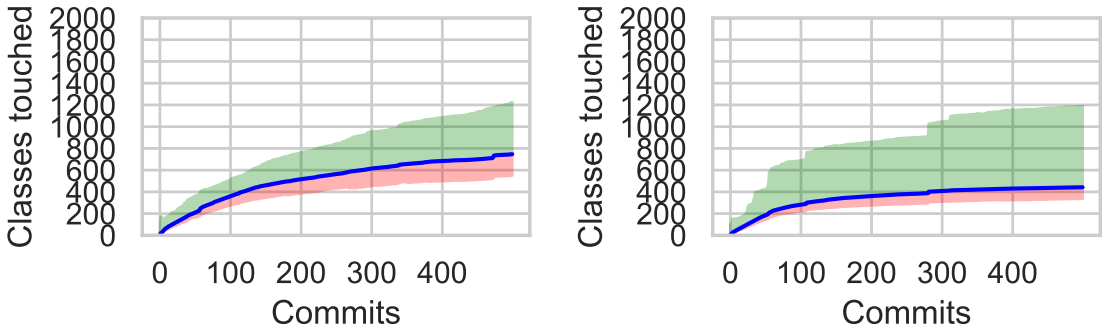
Figure 28: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=36.62, median=18.0 & std=28.13)

(b) Sustained later joiner (n=203, μ=37.08, median=22.0 & std=36.82)

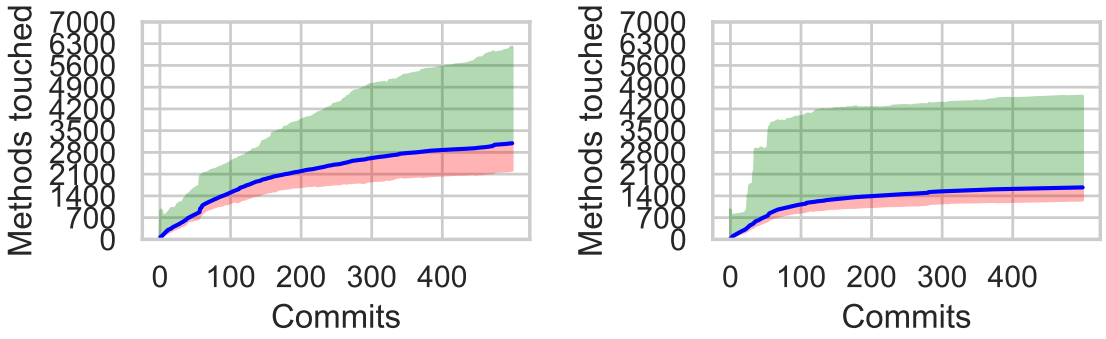
Figure 29: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=746.59, median=426.0 & std=403.07)

(b) Sustained later joiner (n=203, μ=442.33, median=239.0 & std=426.23)

Figure 30: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

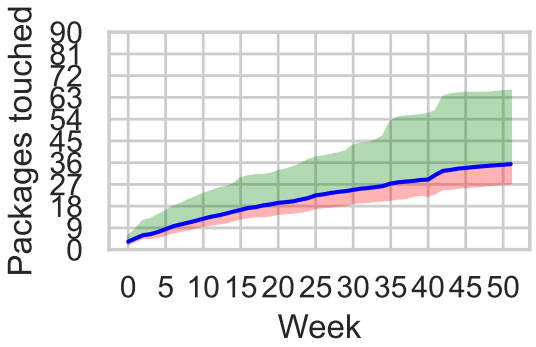


(a) Sustained founder (n=34, μ=3095.18, median=1645.5 & std=2043.47)

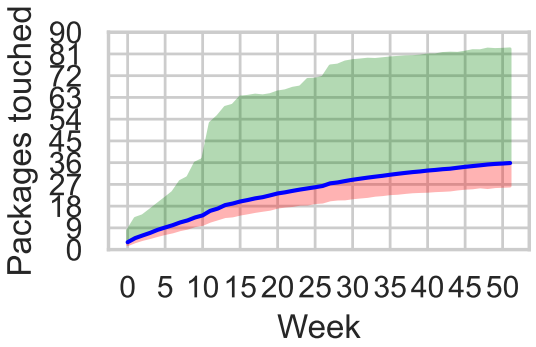
(b) Sustained later joiner (n=203, μ=1672.84, median=838.0 & std=1909.9)

Figure 31: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

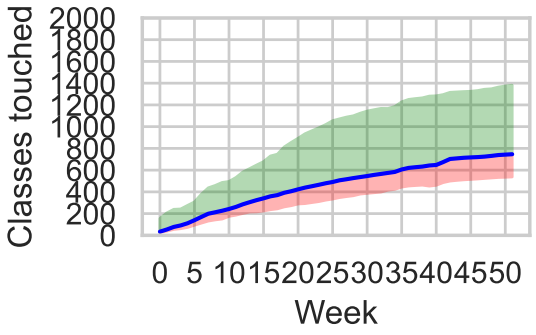
A.5 Time Series Components Touched - For packages touched for each week commit



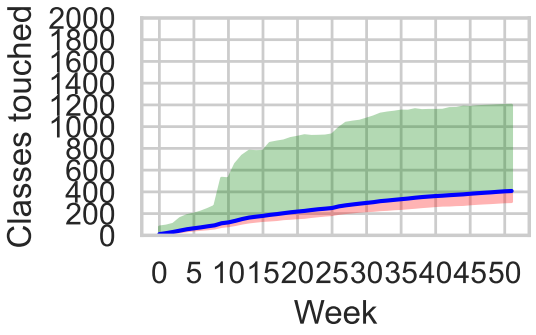
(a) Packages for All founders (n=34, μ=35.38, median=14.0 & std=23.11)



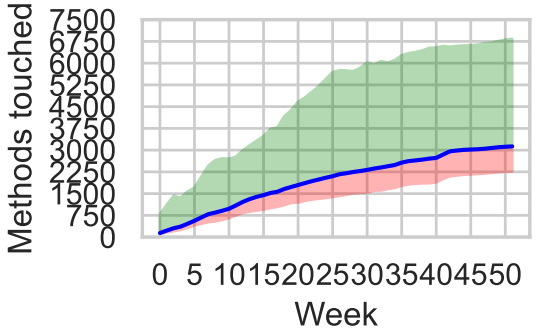
(b) Packages for All joiners (n=203, μ=35.84, median=13.0 & std=33.4)



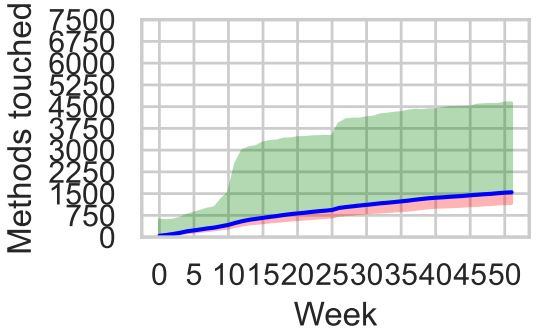
(c) Classes for All founders (n=34, μ=746.15, median=321.28 & std=515.17)



(d) Classes for All joiners (n=203, μ=408.05, median=125.0 & std=440.63)

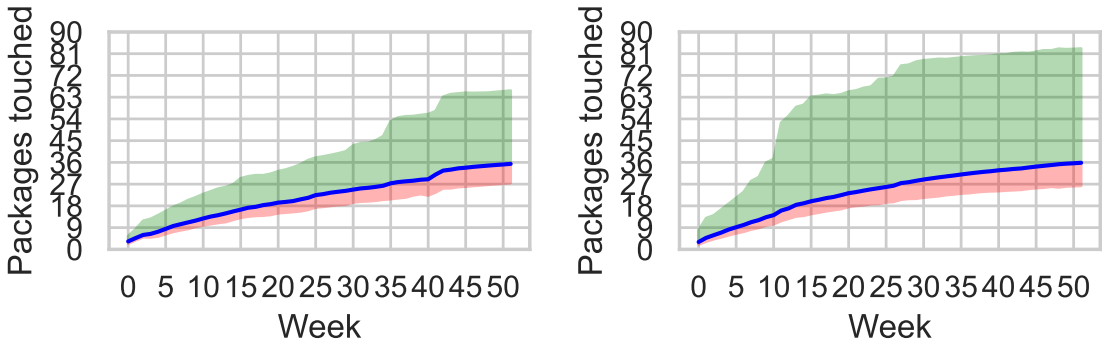


(e) Methods for All founders (n=34, μ=3129.0, median=1296.71 & std=2539.1)



(f) Methods for All joiners (n=203, μ=1545.61, median=435.2 & std=1709.56)

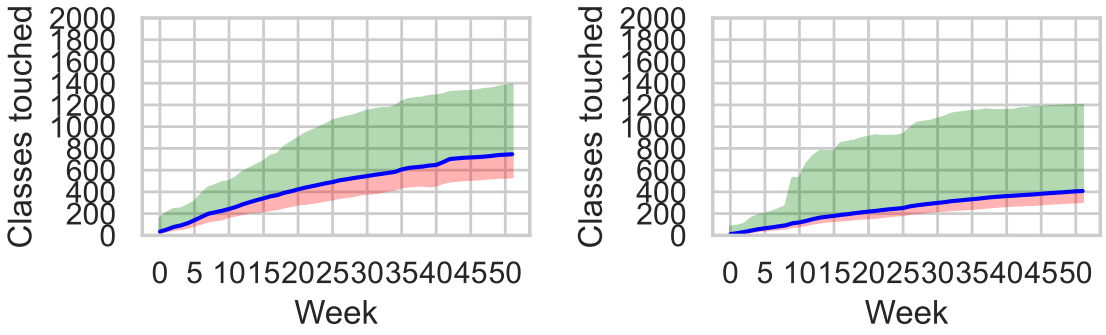
Figure 32: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=35.38, median=14.0 & std=23.11)

(b) Sustained later joiner (n=203, μ=35.84, median=13.0 & std=33.4)

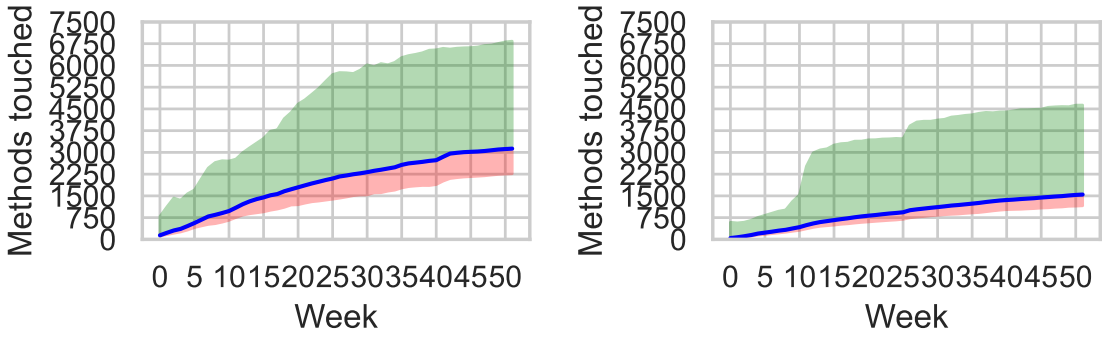
Figure 33: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=746.15, median=321.28 & std=515.17)

(b) Sustained later joiner (n=203, μ=408.05, median=125.0 & std=440.63)

Figure 34: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



(a) Sustained founder (n=34, μ=3129.0, median=1296.71 & std=2539.1)

(b) Sustained later joiner (n=203, μ=1545.61, median=435.2 & std=1709.56)

Figure 35: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

B Statistical Significance Test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances.

Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers.

The Brunner-Münzel test was employed as a robust non-parametric method to determine if the relative treatment effects of Group A differ significantly from Group B. This test is highly appropriate for skewed count data and is completely robust to unequal variances and large group size imbalances. Complementarily, Cliff's δ and the Probability of Superiority (PS) are provided as robust non-parametric effect sizes to quantify the directional dominance magnitude between the samples.

Columns below:

- **Component** - Packages, classes or method touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A. In brackets the number of developers for sample A.
- **Sample B** μ - The mean average of the second sample B. In brackets the number of developers for sample b.
- **Welch Statistic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** P - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU** P - The Mann-Whitney p-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **BM** P - The Brunner-Münzel asymptotic p-value. A $p < 0.05$ indicates a significant difference in relative effects.
- **Brunner-Münzel W** - The robust test statistic. Large absolute values indicate a significant structural shift.
- **Cliff's** δ - Non-parametric effect size bound between -1 and +1, evaluating directional dominance.
- **Prob. of Superiority (PS)** - The empirical probability (0 to 1) that a randomly selected developer from Sample A outranked a developer from Sample B.

Table 25: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of packages after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Founder		Joiner		Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel	Cliff's δ	Prob. Sup. (PS)
			μ	Number	μ	Number		Statistic	P			W		
Packages	1	All	1.54	269	1.22	15948	79.35	3.70	0.0003	0.0000	0.0000	-5.29	0.137	0.569
Packages	1	Moderate	2.55	42	2.59	646	101.84	-0.12	0.9039	0.8829	0.8817	-0.15	0.013	0.506
Packages	1	Sustained	2.21	34	2.40	203	108.98	-0.55	0.5833	0.8805	0.8804	0.15	-0.015	0.492
Packages	2	All	3.08	110	2.97	2978	96.51	0.37	0.7154	0.4243	0.4148	-0.82	0.043	0.522
Packages	2	Moderate	3.48	42	3.58	646	103.00	-0.16	0.8709	0.5758	0.5725	0.57	-0.050	0.475
Packages	2	Sustained	3.00	34	3.57	203	118.88	-1.12	0.2654	0.7985	0.7912	0.27	-0.027	0.487
Packages	5	Transient with five or more; moderate & sustained	4.98	85	6.09	1278	122.44	-2.32	0.0222	0.0425	0.0387	2.10	-0.131	0.435
Packages	5	Moderate	5.26	42	6.40	646	121.59	-1.41	0.1655	0.0674	0.0653	1.89	-0.168	0.416
Packages	5	Sustained	4.74	34	6.13	203	129.41	-1.90	0.0610	0.2268	0.2343	1.21	-0.129	0.436
Packages	10	Moderate & sustained	7.25	75	9.45	783	130.31	-2.62	0.0103	0.0010	0.0017	3.24	-0.230	0.385
Packages	10	Moderate	7.78	41	9.73	580	125.09	-1.54	0.1304	0.0116	0.0180	2.46	-0.235	0.382
Packages	10	Sustained	6.62	34	8.65	203	130.71	-1.85	0.0694	0.1076	0.1319	1.54	-0.172	0.414
Packages	20	Sustained	9.32	34	13.06	203	140.12	-2.31	0.0243	0.0435	0.0572	1.96	-0.216	0.392

Table 26: Statistical significance tests of sample A category versus sample B category touches of different granularities of packages after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Sample				Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			A μ	A Number	B μ	B Number		Statistic	P					
Packages	1	Transient vs. moderate	1.15	15292	2.59	688	225.27	-12.33	0.0000	0.0000	0.0000	27.50	-0.524	0.238
Packages	1	Transient vs. sustained	1.15	15292	2.38	237	206.49	-7.21	0.0000	0.0000	0.0000	14.25	-0.467	0.267
Packages	1	Transient vs. moderate & sustained	1.15	15292	2.54	925	220.46	-14.24	0.0000	0.0000	0.0000	30.82	-0.510	0.245
Packages	1	Moderate vs. sustained	2.59	688	2.38	237	91.66	1.05	0.2945	0.1045	0.1037	-1.63	0.067	0.533
Packages	1	Core vs. Non Core	1.87	198	1.34	16019	71.55	3.87	0.0001	0.0000	0.0000	-5.40	0.186	0.593
Packages	2	Moderate vs. sustained	3.57	688	3.49	237	97.51	0.27	0.7861	0.1265	0.1316	-1.51	0.065	0.533
Packages	2	Core vs. Non Core	3.72	112	2.95	2976	79.23	2.42	0.0168	0.0000	0.0000	-4.63	0.226	0.613
Packages	5	Moderate vs. sustained	6.33	688	5.93	237	93.68	0.90	0.3688	0.0805	0.0802	-1.75	0.076	0.538
Packages	5	Core vs. Non Core	6.21	107	6.01	1256	96.82	0.40	0.6927	0.1946	0.1468	-1.46	0.075	0.538
Packages	10	Moderate vs. sustained	9.60	621	8.36	237	87.03	2.15	0.0320	0.0006	0.0007	-3.43	0.151	0.576
Packages	10	Core vs. Non Core	9.12	104	9.28	754	101.68	-0.21	0.8318	0.8274	0.8149	-0.23	0.013	0.507
Packages	20	Core vs. Non Core	13.00	58	12.37	179	95.19	0.43	0.6665	0.0461	0.0299	-2.19	0.174	0.587

Table 27: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of classes after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Founder		Joiner		Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel	Cliff's	Prob. Sup.
			μ	Number	μ	Number		Statistic	P			W	δ	(PS)
Classes	1	All	2.83	269	1.27	15948	45.03	2.63	0.0090	0.0000	0.0000	-5.15	0.114	0.557
Classes	1	Moderate	7.71	42	2.95	646	38.21	1.41	0.1647	0.4470	0.4887	-0.70	0.055	0.527
Classes	1	Sustained	3.44	34	5.49	203	159.47	-0.80	0.4241	0.1429	0.1814	-1.36	0.117	0.559
Classes	2	All	12.41	110	6.03	2978	48.60	1.70	0.0914	0.2037	0.2607	-1.13	0.069	0.535
Classes	2	Moderate	13.48	42	8.14	646	60.42	1.59	0.1191	0.2599	0.3325	-0.98	0.102	0.551
Classes	2	Sustained	19.62	34	8.25	203	42.04	0.99	0.3269	0.0313	0.0467	-2.05	0.226	0.613
Classes	5	Transient with five or more; moderate & sustained	35.89	85	22.77	1278	63.43	1.78	0.0779	0.0633	0.0800	-1.77	0.120	0.560
Classes	5	Moderate	39.76	42	22.96	646	57.75	1.54	0.1305	0.1476	0.2085	-1.28	0.133	0.567
Classes	5	Sustained	36.32	34	30.01	203	82.62	0.48	0.6350	0.5895	0.5843	-0.55	0.058	0.529
Classes	10	Moderate & sustained	63.96	75	43.51	783	68.03	2.01	0.0483	0.0893	0.1269	-1.54	0.119	0.559
Classes	10	Moderate	60.12	41	41.95	580	69.77	1.32	0.1930	0.2000	0.2654	-1.13	0.120	0.560
Classes	10	Sustained	68.59	34	47.98	203	69.95	1.30	0.2003	0.2478	0.2786	-1.10	0.124	0.562
Classes	20	Sustained	107.50	34	81.91	203	76.20	1.34	0.1863	0.2857	0.3224	-1.00	0.115	0.557

Table 28: Statistical significance tests of sample A category versus sample B category touches of different granularities of classes after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Sample				Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			A μ	A Number	B μ	B Number		Statistic	P					
Classes	1	Transient vs. moderate	1.15	15292	3.24	688	281.11	-6.43	0.0000	0.0000	0.0000	14.28	-0.244	0.378
Classes	1	Transient vs. sustained	1.15	15292	5.19	237	450.89	-2.07	0.0398	0.0000	0.0000	7.54	-0.209	0.395
Classes	1	Transient vs. moderate & sustained	1.15	15292	3.74	925	324.61	-4.65	0.0000	0.0000	0.0000	16.12	-0.235	0.382
Classes	1	Moderate vs. sustained	3.24	688	5.19	237	160.39	-0.99	0.3247	0.2623	0.2499	-1.15	0.038	0.519
Classes	1	Core vs. Non Core	3.67	198	1.28	16019	34.99	2.27	0.0242	0.0000	0.0001	-3.99	0.100	0.550
Classes	2	Moderate vs. sustained	8.47	688	9.88	237	116.65	-0.56	0.5754	0.3370	0.3344	-0.97	0.041	0.521
Classes	2	Core vs. Non Core	12.88	112	6.01	2976	46.64	1.77	0.0802	0.0084	0.0105	-2.60	0.142	0.571
Classes	5	Moderate vs. sustained	23.99	688	30.92	237	128.88	-1.54	0.1256	0.6877	0.6889	-0.40	0.017	0.509
Classes	5	Core vs. Non Core	30.73	107	22.98	1256	74.78	1.24	0.2161	0.7104	0.7285	0.35	-0.022	0.489
Classes	10	Moderate vs. sustained	43.15	621	50.94	237	118.05	-1.53	0.1282	0.6770	0.6812	-0.41	0.018	0.509
Classes	10	Core vs. Non Core	54.99	104	43.96	754	79.95	1.38	0.1697	0.9853	0.9861	-0.02	0.001	0.501
Classes	20	Core vs. Non Core	104.21	58	79.55	179	76.34	1.94	0.0557	0.0014	0.0011	-3.36	0.280	0.640

Table 29: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of methods after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Founder		Joiner		Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			μ	Number	μ	Number		Statistic	P					
Methods	1	All	7.16	269	2.01	15948	28.08	2.37	0.0185	0.0000	0.0000	-5.38	0.123	0.562
Methods	1	Moderate	23.71	42	7.92	646	33.42	1.28	0.2075	0.5319	0.5660	-0.58	0.047	0.524
Methods	1	Sustained	10.59	34	24.08	203	227.41	-0.78	0.4382	0.2351	0.2715	-1.11	0.098	0.549
Methods	2	All	54.12	110	19.93	2978	36.84	1.71	0.0893	0.3344	0.4015	-0.84	0.053	0.527
Methods	2	Moderate	61.76	42	29.98	646	48.54	1.60	0.1163	0.3543	0.4320	-0.79	0.085	0.542
Methods	2	Sustained	88.32	34	25.84	203	29.26	1.04	0.3041	0.0476	0.0667	-1.88	0.211	0.605
Methods	5	Transient with five or more; moderate & sustained	149.40	85	80.32	1278	53.76	1.59	0.1148	0.0648	0.0836	-1.75	0.119	0.560
Methods	5	Moderate	179.29	42	80.93	646	45.14	1.37	0.1778	0.1127	0.1635	-1.42	0.146	0.573
Methods	5	Sustained	138.53	34	114.77	203	82.85	0.36	0.7211	0.7890	0.7946	-0.26	0.029	0.514
Methods	10	Moderate & sustained	259.76	75	154.41	783	59.44	1.72	0.0890	0.0583	0.0920	-1.70	0.132	0.566
Methods	10	Moderate	264.83	41	145.21	580	54.83	1.25	0.2202	0.1690	0.2351	-1.20	0.128	0.564
Methods	10	Sustained	253.65	34	180.70	203	71.24	1.00	0.3240	0.2291	0.2666	-1.13	0.129	0.565
Methods	20	Sustained	401.24	34	295.10	203	73.55	1.16	0.2533	0.5131	0.5561	-0.59	0.070	0.535

Table 30: Statistical significance tests of sample A category versus sample B category touches of different granularities of methods after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First n commits	Contributor Category	Sample				Percentage Difference in μ	Welch		MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
			A μ	A Number	B μ	B Number		Statistic	P					
Methods	1	Transient vs. moderate	1.48	15292	8.89	688	600.13	-5.74	0.0000	0.0000	0.0000	15.67	-0.278	0.361
Methods	1	Transient vs. sustained	1.48	15292	22.14	237	1495.13	-1.46	0.1460	0.0000	0.0000	7.93	-0.227	0.386
Methods	1	Transient vs. moderate & sustained	1.48	15292	12.28	925	829.44	-2.88	0.0041	0.0000	0.0000	17.51	-0.265	0.367
Methods	1	Moderate vs. sustained	8.89	688	22.14	237	249.14	-0.93	0.3523	0.1376	0.1271	-1.53	0.052	0.526
Methods	1	Core vs. Non Core	9.51	198	2.05	16019	21.56	1.90	0.0588	0.0000	0.0000	-4.18	0.109	0.555
Methods	2	Moderate vs. sustained	31.92	688	34.81	237	109.05	-0.26	0.7956	0.4271	0.4155	-0.82	0.034	0.517
Methods	2	Core vs. Non Core	52.45	112	19.97	2976	38.09	1.62	0.1072	0.0034	0.0032	-3.01	0.160	0.580
Methods	5	Moderate vs. sustained	86.94	688	118.18	237	135.93	-1.36	0.1747	0.8232	0.8227	-0.22	0.010	0.505
Methods	5	Core vs. Non Core	119.94	107	81.62	1256	68.05	1.08	0.2847	0.3926	0.4154	0.82	-0.050	0.475
Methods	10	Moderate vs. sustained	153.10	621	191.16	237	124.86	-1.50	0.1340	0.9083	0.9101	0.11	-0.005	0.497
Methods	10	Core vs. Non Core	210.71	104	157.12	754	74.57	1.14	0.2572	0.2740	0.3184	1.00	-0.066	0.467
Methods	20	Core vs. Non Core	347.88	58	298.16	179	85.71	0.83	0.4113	0.1034	0.1000	-1.66	0.142	0.571

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{8}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} \tag{9}$$

Mann-Whitney U Test To evaluate the differences in the stochastic dominance of architectural component exploration between primary roles (**Sample A** vs. **Sample B**) without assuming a normal distribution, the non-parametric Mann-Whitney U test (Wilcoxon rank-sum test) is employed.

The data from both independent groups are combined and ranked globally from lowest (1) to highest ($N_1 + N_2$), resolving ties by assigning the average rank. The U statistic for Sample A (U_1) is calculated as:

$$U_1 = R_1 - \frac{N_1(N_1 + 1)}{2} \tag{10}$$

Where:

- N_1 is the sample size of Sample A.
- R_1 is the sum of the ranks assigned to Sample A.
- The total number of pairwise comparisons (maximum possible pairs) is defined by $N_1 \times N_2$.

Normal Approximation for Large Samples Given the substantial sample sizes present in repository participant metrics, the distribution of the U statistic rapidly approaches a normal distribution. The asymptotic significance is determined using a standard normal Z-score approximation:

$$Z = \frac{U_1 - \mu_U}{\sigma_U} \tag{11}$$

Where the expected mean (μ_U) and standard deviation (σ_U) under the null hypothesis of identical distributions are defined as:

$$\mu_U = \frac{N_1 N_2}{2}, \quad \sigma_U = \sqrt{\frac{N_1 N_2 (N_1 + N_2 + 1)}{12}} \tag{12}$$

A calculated p-value lower than the alpha threshold ($p < 0.05$) indicates a statistically significant difference in the rank-order distributions of the two groups, demonstrating that one developer role systematically outranks the other in codebase exploration.

Brunner-Münzel Test To test for differences in the stochastic distributions of primary roles (**Sample A** vs. **Sample B**) without assuming normal distribution or variance homogeneity, the Brunner-Münzel test is applied. The test statistic W is defined as:

$$W = \sqrt{\frac{N_1 N_2}{N_1 + N_2}} \cdot \frac{\hat{p} - \frac{1}{2}}{\sqrt{\hat{\sigma}^2}} \tag{13}$$

Where $\hat{p}$ represents the relative treatment effect (Probability of Superiority), reflecting the probability that a random observation from Sample A is larger than one from Sample B plus half the probability that they are equal.

Cliff's δ Effect Size To complement hypothesis significance testing with a distribution-free magnitude metric, Cliff's δ is tracked to express the degree of dominance of one sample over the other:

$$\delta = \frac{\sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{sign}(X_{1i} - X_{2j})}{N_1 N_2} \tag{14}$$

Where the sign function yields $+1$ when an element of Sample A is larger than Sample B, 0 if equal, and -1 if smaller. Effect sizes are interpreted using standard empirical thresholds ($|\delta| < 0.147$ is negligible, ≥ 0.147 is small, ≥ 0.330 is medium, and ≥ 0.474 is large).

C Standard Anova - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

Table 31: Anova results for packages for number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
packages	1	16217	Role (F/LJ)	0.09	1	0.07	0.7976	0.0000
packages	1	16217	Category (T/M/S)	79.08	2	30.11	0.0000	0.0037
packages	1	16217	Interaction	1.82	2	0.69	0.5007	0.0001
packages	1	16217	Residual	21288.02	16211	-	-	
packages	2	3088	Role (F/LJ)	0.43	1	0.04	0.8354	0.0000
packages	2	3088	Category (M/S)	12.35	2	0.62	0.5375	0.0004
packages	2	3088	Interaction	4.91	2	0.25	0.7812	0.0002
packages	2	3088	Residual	30641.08	3082	-	-	
packages	5	1363	Role (F/LJ)	50.89	1	1.49	0.2227	0.0011
packages	5	1363	Category (M/S)	6.99	2	0.10	0.9028	0.0001
packages	5	1363	Interaction	1.37	2	0.02	0.9802	0.0000
packages	5	1363	Residual	46399.69	1357	-	-	
packages	10	858	Role (F/LJ)	145.95	1	2.65	0.1036	0.0029
packages	10	858	Category (M/S)	25.13	1	0.46	0.4992	0.0005
packages	10	858	Interaction	0.11	1	0.00	0.9649	0.0000
packages	10	858	Residual	46952.80	854	-	-	

Table 32: Anova results for classes for number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
classes	1	16217	Role (F/LJ)	896.13	1	45.17	0.0000	0.0027
classes	1	16217	Category (T/M/S)	1279.75	2	32.25	0.0000	0.0039
classes	1	16217	Interaction	870.94	2	21.95	0.0000	0.0027
classes	1	16217	Residual	321640.64	16211	-	-	
classes	2	3088	Role (F/LJ)	1121.92	1	4.65	0.0312	0.0015
classes	2	3088	Category (M/S)	4286.56	2	8.88	0.0001	0.0056
classes	2	3088	Interaction	2511.99	2	5.20	0.0056	0.0033
classes	2	3088	Residual	744173.43	3082	-	-	
classes	5	1363	Role (F/LJ)	11129.18	1	7.32	0.0069	0.0052
classes	5	1363	Category (M/S)	4117.43	2	1.35	0.2587	0.0019
classes	5	1363	Interaction	3640.20	2	1.20	0.3025	0.0017
classes	5	1363	Residual	2063860.58	1357	-	-	
classes	10	858	Role (F/LJ)	12645.13	1	4.39	0.0365	0.0048
classes	10	858	Category (M/S)	1332.25	1	0.46	0.4967	0.0005
classes	10	858	Interaction	98.16	1	0.03	0.8536	0.0000
classes	10	858	Residual	2460690.10	854	-	-	

Table 33: Anova results for methods for number of commits and categories of developers

Component	First n commits	Number of Developers	Factor	Sum Squared	df	F-Stat	P Value	ETA Sq
methods	1	16217	Role (F/LJ)	9832.53	1	11.96	0.0005	0.0007
methods	1	16217	Category (T/M/S)	15316.33	2	9.31	0.0001	0.0011
methods	1	16217	Interaction	14535.91	2	8.84	0.0001	0.0011
methods	1	16217	Residual	13328717.62	16211	-	-	
methods	2	3088	Role (F/LJ)	39842.01	1	7.77	0.0053	0.0025
methods	2	3088	Category (M/S)	107007.93	2	10.44	0.0000	0.0066
methods	2	3088	Interaction	73539.47	2	7.17	0.0008	0.0045
methods	2	3088	Residual	15801249.51	3082	-	-	
methods	5	1363	Role (F/LJ)	381459.60	1	9.72	0.0019	0.0069
methods	5	1363	Category (M/S)	128673.36	2	1.64	0.1946	0.0023
methods	5	1363	Interaction	141668.49	2	1.80	0.1650	0.0026
methods	5	1363	Residual	53273281.50	1357	-	-	
methods	10	858	Role (F/LJ)	547971.01	1	6.98	0.0084	0.0078
methods	10	858	Category (M/S)	2324.11	1	0.03	0.8634	0.0000
methods	10	858	Interaction	36040.01	1	0.46	0.4981	0.0005
methods	10	858	Residual	66999194.82	854	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{15}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{16}$$

D Tukey HSD - For all components touched for all commits

A Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 34: Significant Tukey HSD results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16217	Founder Moderate	Founder Sustained	-4.27	0.0005
classes	1	16217	Founder Moderate	Founder Transient	-6.06	0.0000
classes	1	16217	Founder Moderate	Later Joiner Moderate	-4.77	0.0000
classes	1	16217	Founder Moderate	Later Joiner Sustained	-2.23	0.0376
classes	1	16217	Founder Moderate	Later Joiner Transient	-6.57	0.0000
classes	1	16217	Founder Sustained	Later Joiner Transient	-2.30	0.0321
classes	1	16217	Founder Transient	Later Joiner Moderate	1.29	0.0056
classes	1	16217	Founder Transient	Later Joiner Sustained	3.83	0.0000
classes	1	16217	Later Joiner Moderate	Later Joiner Sustained	2.54	0.0000
classes	1	16217	Later Joiner Moderate	Later Joiner Transient	-1.80	0.0000
classes	1	16217	Later Joiner Sustained	Later Joiner Transient	-4.34	0.0000
classes	2	3088	Founder Moderate	Later Joiner Transient	-8.30	0.0081
classes	2	3088	Founder Sustained	Founder Transient	-15.74	0.0004
classes	2	3088	Founder Sustained	Later Joiner Moderate	-11.48	0.0004
classes	2	3088	Founder Sustained	Later Joiner Sustained	-11.37	0.0011
classes	2	3088	Founder Sustained	Later Joiner Transient	-14.44	0.0000
classes	2	3088	Later Joiner Moderate	Later Joiner Transient	-2.96	0.0003
classes	5	1363	Founder Moderate	Later Joiner Transient	-20.71	0.0134
classes	5	1363	Later Joiner Sustained	Later Joiner Transient	-10.96	0.0128
classes	10	858	Founder Sustained	Later Joiner Moderate	-26.64	0.0258

Table 35: Significant Tukey HSD results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16217	Founder Moderate	Founder Transient	-20.76	0.0003
methods	1	16217	Founder Moderate	Later Joiner Moderate	-15.79	0.0072
methods	1	16217	Founder Moderate	Later Joiner Transient	-22.25	0.0000
methods	1	16217	Founder Transient	Later Joiner Sustained	21.12	0.0000
methods	1	16217	Later Joiner Moderate	Later Joiner Sustained	16.15	0.0000
methods	1	16217	Later Joiner Moderate	Later Joiner Transient	-6.46	0.0000
methods	1	16217	Later Joiner Sustained	Later Joiner Transient	-22.62	0.0000
methods	2	3088	Founder Moderate	Founder Transient	-51.29	0.0236
methods	2	3088	Founder Moderate	Later Joiner Sustained	-35.92	0.0367
methods	2	3088	Founder Moderate	Later Joiner Transient	-45.44	0.0007
methods	2	3088	Founder Sustained	Founder Transient	-77.85	0.0001
methods	2	3088	Founder Sustained	Later Joiner Moderate	-58.35	0.0001
methods	2	3088	Founder Sustained	Later Joiner Sustained	-62.48	0.0000
methods	2	3088	Founder Sustained	Later Joiner Transient	-72.00	0.0000
methods	2	3088	Later Joiner Moderate	Later Joiner Transient	-13.65	0.0003
methods	5	1363	Founder Moderate	Later Joiner Moderate	-98.35	0.0229
methods	5	1363	Founder Moderate	Later Joiner Transient	-116.18	0.0040
methods	5	1363	Later Joiner Sustained	Later Joiner Transient	-51.66	0.0273
methods	10	858	Founder Moderate	Later Joiner Moderate	-119.62	0.0416

Table 36: Not Significant Tukey HSD results for classes and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
classes	1	16217	Founder Sustained	Founder Transient	-1.78	0.2606
classes	1	16217	Founder Sustained	Later Joiner Moderate	-0.49	0.9888
classes	1	16217	Founder Sustained	Later Joiner Sustained	2.05	0.1302
classes	1	16217	Founder Transient	Later Joiner Transient	-0.51	0.6063
classes	2	3088	Founder Moderate	Founder Sustained	6.14	0.5230
classes	2	3088	Founder Moderate	Founder Transient	-9.59	0.0803
classes	2	3088	Founder Moderate	Later Joiner Moderate	-5.33	0.2592
classes	2	3088	Founder Moderate	Later Joiner Sustained	-5.23	0.3508
classes	2	3088	Founder Transient	Later Joiner Moderate	4.26	0.6264
classes	2	3088	Founder Transient	Later Joiner Sustained	4.36	0.6542
classes	2	3088	Founder Transient	Later Joiner Transient	1.30	0.9968
classes	2	3088	Later Joiner Moderate	Later Joiner Sustained	0.10	1.0000
classes	2	3088	Later Joiner Sustained	Later Joiner Transient	-3.07	0.0780
classes	5	1363	Founder Moderate	Founder Sustained	-3.44	0.9989
classes	5	1363	Founder Moderate	Founder Transient	-23.54	0.5698
classes	5	1363	Founder Moderate	Later Joiner Moderate	-16.80	0.0749
classes	5	1363	Founder Moderate	Later Joiner Sustained	-9.75	0.6803
classes	5	1363	Founder Sustained	Founder Transient	-20.10	0.7421
classes	5	1363	Founder Sustained	Later Joiner Moderate	-13.36	0.3738
classes	5	1363	Founder Sustained	Later Joiner Sustained	-6.31	0.9528
classes	5	1363	Founder Sustained	Later Joiner Transient	-17.27	0.1289
classes	5	1363	Founder Transient	Later Joiner Moderate	6.74	0.9956
classes	5	1363	Founder Transient	Later Joiner Sustained	13.79	0.9051
classes	5	1363	Founder Transient	Later Joiner Transient	2.83	0.9999
classes	5	1363	Later Joiner Moderate	Later Joiner Sustained	7.05	0.2176
classes	5	1363	Later Joiner Moderate	Later Joiner Transient	-3.91	0.5917
classes	10	858	Founder Moderate	Founder Sustained	8.47	0.9047
classes	10	858	Founder Moderate	Later Joiner Moderate	-18.17	0.1555
classes	10	858	Founder Moderate	Later Joiner Sustained	-12.14	0.5497
classes	10	858	Founder Sustained	Later Joiner Sustained	-20.61	0.1632
classes	10	858	Later Joiner Moderate	Later Joiner Sustained	6.03	0.5138
classes	20	237	Founder Sustained	Later Joiner Sustained	-25.59	0.1034

Table 37: Not Significant Tukey HSD results for methods and number of commits

Component	Commits	Number of Developers	Group		Mean Diff	Adj P Value
			A	B		
methods	1	16217	Founder Moderate	Founder Sustained	-13.13	0.3513
methods	1	16217	Founder Moderate	Later Joiner Sustained	0.36	1.0000
methods	1	16217	Founder Sustained	Founder Transient	-7.63	0.7083
methods	1	16217	Founder Sustained	Later Joiner Moderate	-2.66	0.9951
methods	1	16217	Founder Sustained	Later Joiner Sustained	13.49	0.1130
methods	1	16217	Founder Sustained	Later Joiner Transient	-9.13	0.4311
methods	1	16217	Founder Transient	Later Joiner Moderate	4.97	0.2814
methods	1	16217	Founder Transient	Later Joiner Transient	-1.50	0.9795
methods	2	3088	Founder Moderate	Founder Sustained	26.56	0.5933
methods	2	3088	Founder Moderate	Later Joiner Moderate	-31.79	0.0597
methods	2	3088	Founder Transient	Later Joiner Moderate	19.51	0.6329
methods	2	3088	Founder Transient	Later Joiner Sustained	15.37	0.8564
methods	2	3088	Founder Transient	Later Joiner Transient	5.85	0.9971
methods	2	3088	Later Joiner Moderate	Later Joiner Sustained	-4.13	0.9799
methods	2	3088	Later Joiner Sustained	Later Joiner Transient	-9.52	0.4596
methods	5	1363	Founder Moderate	Founder Sustained	-40.76	0.9486
methods	5	1363	Founder Moderate	Founder Transient	-128.29	0.4904
methods	5	1363	Founder Moderate	Later Joiner Sustained	-64.52	0.3897
methods	5	1363	Founder Sustained	Founder Transient	-87.53	0.8472
methods	5	1363	Founder Sustained	Later Joiner Moderate	-57.59	0.5640
methods	5	1363	Founder Sustained	Later Joiner Sustained	-23.76	0.9873
methods	5	1363	Founder Sustained	Later Joiner Transient	-75.42	0.2690
methods	5	1363	Founder Transient	Later Joiner Moderate	29.93	0.9977
methods	5	1363	Founder Transient	Later Joiner Sustained	63.77	0.9347
methods	5	1363	Founder Transient	Later Joiner Transient	12.11	1.0000
methods	5	1363	Later Joiner Moderate	Later Joiner Sustained	33.83	0.2763
methods	5	1363	Later Joiner Moderate	Later Joiner Transient	-17.83	0.6995
methods	10	858	Founder Moderate	Founder Sustained	-11.18	0.9982
methods	10	858	Founder Moderate	Later Joiner Sustained	-84.13	0.2964
methods	10	858	Founder Sustained	Later Joiner Moderate	-108.44	0.1256
methods	10	858	Founder Sustained	Later Joiner Sustained	-72.95	0.4962
methods	10	858	Later Joiner Moderate	Later Joiner Sustained	35.49	0.4058
methods	20	237	Founder Sustained	Later Joiner Sustained	-106.14	0.1554

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{17}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df}\sqrt{\frac{MS_{residual}}{n}} \tag{18}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

E Seaborn Stripplot - Number of Developers versus Packages Touched

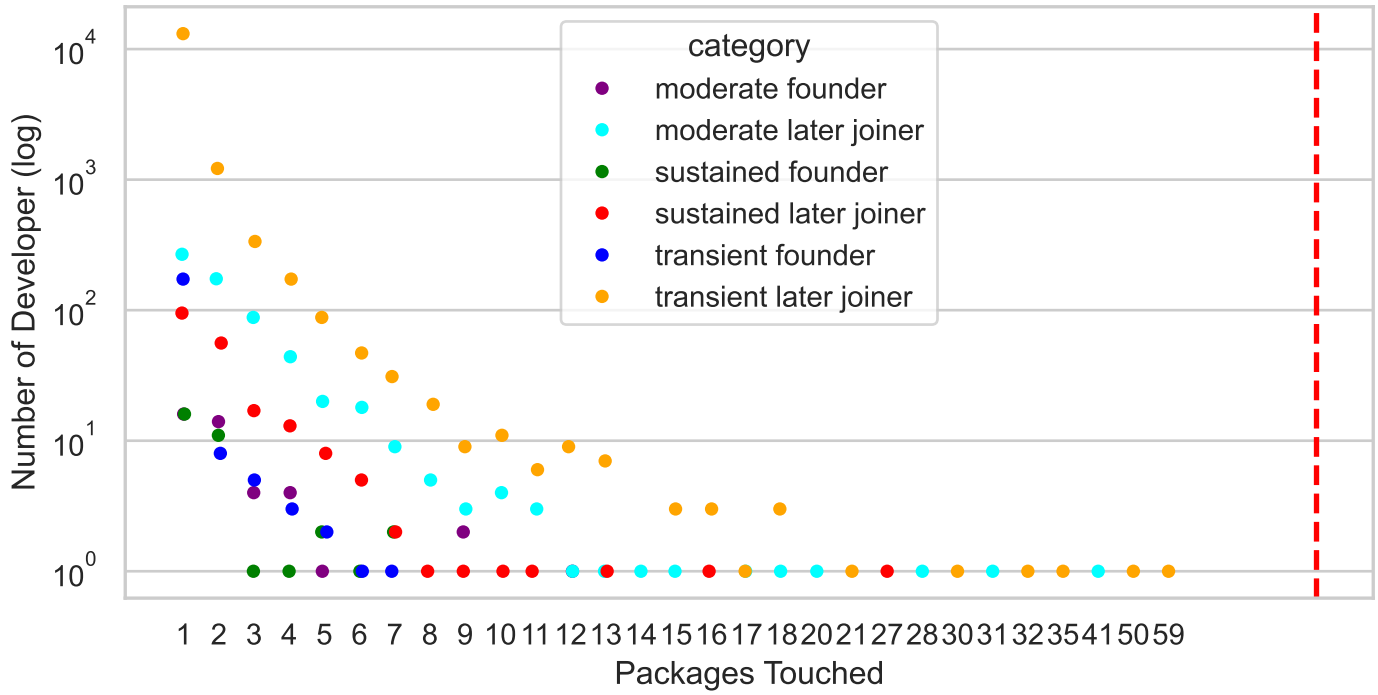


Figure 36: A seaborn strip plot of packages touched (y-axis) against number of developers (x-axis) made by from six (6) categories: Transient Founder (Blue, 193), Moderate Founder (Purple, 42), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 15099), Moderate Later Joiner (Cyan, 646), Sustained Later Joiner (Red, 203).